

# Quantum-to-classical Lax correspondence in non-Hermitian XXX deformations

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**Abstract.** We study the quantum-to-classical Lax correspondence for the Class 5 and Class 6 non-Hermitian deformations of the Heisenberg XXX chain. Starting from the finite-lattice Lax equation, we take a fixed-spin- $\frac{1}{2}$  coherent-state long-wavelength limit. The spatial component follows from the leading lower symbol, but the time component is subtler because the two products in the local lattice zero-curvature equation overlap on a physical site and their lower symbols do not factorize. We show that the leading shared-site contributions cancel, while the next spatial Taylor and operator-product terms combine, through the Sutherland relation, into a local correction governed by the second moment of the leading spatial Lax coefficient. The resulting continuum zero-curvature equations are off-shell equivalent to the independently derived Hamiltonian equations. In Class 5 the source is a total spatial derivative, whereas in Class 6 it contains a nonzero commutator. The continuum Lax pairs are related to the null-like warped  $SL(2)$  and rational eleven-vertex Landau–Lifshitz representations, and the undeformed XXX limit provides a nontrivial check.

**Keywords:** Integrable spin chains; Landau–Lifshitz models; Lax pairs; coherent states

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# 1 Introduction

Landau–Lifshitz models arise as long-wavelength descriptions of quantum spin chains and provide a standard setting in which quantum and classical integrability can be compared. For the Heisenberg chain, the coherent-state continuum limit also reproduces the effective action obtained from strings with large angular momentum [1]. Integrable deformations lead to anisotropic Landau–Lifshitz models and connect spin-chain deformations with deformed string sigma models [2].

The continuum limit of an integrable lattice model involves more than its Hamiltonian. Coherent-state limits of lattice monodromies and spatial Lax operators were developed systematically in Ref. [3], while continuum Hamiltonians, time Lax matrices and boundary terms were studied in Ref. [4]. On the quantum lattice, the time component can be generated from the monodromy and local Hamiltonians [5]. In global lattice sums, coincident site labels can be subleading because they reduce the number of independent sums [3]. A local lattice zero-curvature equation has a different structure: its neighbouring factors can overlap on a physical site, so the corresponding operator products must be treated before taking a coherent-state continuum limit.

The Class 5 and Class 6 chains are two non-Hermitian spin- $\frac{1}{2}$  deformations of the Heisenberg XXX chain that appear in the classification of regular nearest-neighbour integrable Hamiltonians [6]. Their transfer matrices can have nontrivial Jordan blocks. The associated Landau–Lifshitz continuum models and higher conserved charges were derived in Ref. [7]. In Class 5 the deformation is related to a Jordanian Drinfeld twist. The same work constructed a classical spatial Lax matrix from the Class 5 quantum Lax operator and established its quadratic Poisson algebra, while a compatible time component within that construction was left open.

The corresponding classical dynamics are also related to known integrable systems. A null-like warped  $SL(2)$  Landau–Lifshitz model with a classical Lax representation was studied in Ref. [2]. Class 6 is the eleven-vertex model at the quantum level [8], and the corresponding rational Landau–Lifshitz hierarchy is known [9, 10]. These known classical representations provide reference points for studying how the continuum time component emerges directly from the quantum lattice Lax structure.

At fixed spin  $\frac{1}{2}$ , the quantum-to-classical passage requires care with operator products that share a physical site. Smoothness suppresses differences between neighbouring coherent-state labels. Products acting on a common site retain the finite-spin operator algebra. This nonmultiplicativity is standard in coherent-state symbol calculus [11, 12], and finite-spin star products provide an explicit description of such products [13]. The relevant question for the local lattice zero-curvature equation is therefore which shared-site terms survive the long-wavelength expansion and whether they can be incorporated into a local continuum Lax pair.

In this paper we derive the continuum Lax correspondence for the Class 5 and Class 6 chains directly from their finite-lattice Lax equations. The leading shared-site contributions cancel, while the first spatial Taylor terms and the next operator-product terms survive at the order of the continuum time evolution. Using the corresponding order of the Sutherland relation, these terms combine into a local correction determined by the second moment of the leading spatial Lax coefficient. In Class 5 the resulting source is a total spatial derivative, whereas in Class 6 it contains a nonzero commutator. For both models the corrected zero-curvature equations are equivalent off shell to the independently derived Hamiltonian equations. The Class 5 result agrees, up to the appropriate local transformations, with both an independent classical  $r$ -matrix construction and the null-like warped  $SL(2)$  Landau–Lifshitz representation, while the Class 6 result maps to the rational eleven-vertex pair. The undeformed XXX limit provides a further check: neglecting the shared-site contribution gives an incorrect coefficient in the standard time matrix.

Numerical coefficient searches were used during the construction of the two pairs, following the optimization approach of Ref. [14]. The final statements are established analytically; the numerical ansätze, losses and checks are included as supporting material in the appendix.

Section 2 fixes the lattice and continuum conventions and derives the shared-site contribution. Sections 3 and 4 treat Class 5 and Class 6, respectively. Section 5 compares their continuum corrections and the XXX limit. Section 6 summarizes the results and their scope.

## 2 Quantum lattice setup and continuum limit

We fix the lattice Lax equation, coherent-state lower symbols and continuum scaling before deriving the shared-site contribution to the time Lax matrix.

### 2.1 Quantum spin chains and the lattice Lax equation

The two lattice models studied in this paper are homogeneous periodic spin chains with nearest-neighbour interactions and a two-dimensional local Hilbert space. We use the standard quantum inverse-scattering framework for such chains [15, 3], with the following notation.

Let  $\mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2$  denote a reference local Hilbert space. Consider a chain with  $N$  lattice sites, and let  $\mathcal{H}_n$  be a copy of  $\mathcal{H}_{\text{loc}}$  assigned to the  $n$ th site,

$$\mathcal{H}_n \simeq \mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2, \quad n = 1, \dots, N. \quad (2.1)$$

The Hilbert space of the full chain is the tensor product

$$\mathcal{H}^{(N)} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \dots \otimes \mathcal{H}_N. \quad (2.2)$$

We impose periodic boundary conditions, so the site labels are understood modulo  $N$ ; in particular, the site following  $N$  is site 1.

A homogeneous nearest-neighbour Hamiltonian is specified by a single *two-site Hamiltonian operator*

$$h \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (2.3)$$

The adjective “homogeneous” means that the same two-site operator  $h$  is used on every nearest-neighbour bond. We denote by  $h_{n,n+1}$  the operator on the full Hilbert space (2.2) that acts as  $h$  on  $\mathcal{H}_n \otimes \mathcal{H}_{n+1}$  and as the identity on all remaining sites. For  $1 \leq n < N$ , this embedding is explicitly

$$h_{n,n+1} = \mathbf{1}_{\mathcal{H}_1} \otimes \dots \otimes \mathbf{1}_{\mathcal{H}_{n-1}} \otimes h \otimes \mathbf{1}_{\mathcal{H}_{n+2}} \otimes \dots \otimes \mathbf{1}_{\mathcal{H}_N}. \quad (2.4)$$

For the bond joining the last and first sites,  $h_{N,1}$  denotes the same operator  $h$  acting on  $\mathcal{H}_N \otimes \mathcal{H}_1$ , with the other sites left unchanged. Thus  $h_{n,n+1}$  is the copy of the fixed two-site operator  $h$  on the  $n$ th bond.

With these conventions, the Hamiltonian of the periodic chain is

$$H^{\text{lat}} = \sum_{n=1}^N h_{n,n+1}, \quad h_{N,N+1} \equiv h_{N,1}. \quad (2.5)$$

The model dependence therefore enters through the choice of the single matrix  $h$ . We next introduce the quantum  $R$ -matrix and the lattice Lax equation associated with a nearest-neighbour integrable chain.

For the integrable chains considered in this paper, the local interaction is encoded by a quantum  $R$ -matrix. Introduce an auxiliary space

$$\mathcal{V}_a \simeq \mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2, \quad (2.6)$$

which is a tensor factor distinct from the physical Hilbert spaces  $\mathcal{H}_n$ . When several auxiliary spaces are required, we label them by  $a, b, c$ .

Let  $u \in \mathbb{C}$  denote the *quantum spectral parameter*. A quantum  $R$ -matrix in the representation used here is a family of two-site operators

$$R(u) \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (2.7)$$

Whenever two tensor factors are individually labelled, the corresponding subscripts indicate the copy of this same two-site operator acting on those factors. Thus, for example,  $R_{ab}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{V}_b)$  and  $R_{a,n}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{H}_n)$ .

The models considered here have an  $R$ -matrix of difference form. Its Yang–Baxter equation is

$$\begin{aligned} R_{ab}(u-v) R_{ac}(u-w) R_{bc}(v-w) \\ = R_{bc}(v-w) R_{ac}(u-w) R_{ab}(u-v), \end{aligned} \quad (2.8)$$

which is an operator identity on  $\mathcal{V}_a \otimes \mathcal{V}_b \otimes \mathcal{V}_c$  [15]. The subscripts indicate the two tensor factors on which each copy of  $R$  acts.

Let

$$P \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}) \quad (2.9)$$

be the permutation operator defined by  $P(|v\rangle \otimes |w\rangle) = |w\rangle \otimes |v\rangle$ . As for the  $R$ -matrix,  $P_{ab}$  denotes its copy on  $\mathcal{V}_a \otimes \mathcal{V}_b$ . We normalize the  $R$ -matrix so that its regular point is  $u = 0$ ,

$$R(0) = P, \quad \text{equivalently} \quad R_{ab}(0) = P_{ab}. \quad (2.10)$$

For a regular fundamental spin- $\frac{1}{2}$  chain, the two-site Hamiltonian operator introduced in (2.3) is obtained from the same two-site  $R$ -matrix as

$$h = P \left. \frac{\partial R(u)}{\partial u} \right|_{u=0}. \quad (2.11)$$

This is the standard relation between a regular  $R$ -matrix and the nearest-neighbour Hamiltonian density in the quantum inverse-scattering construction [15, 6].

A scalar rescaling  $R(u) \mapsto f(u)R(u)$  with  $f(0) = 1$  preserves the regularity convention (2.10) and adds  $f'(0)\mathbf{1}$  to  $h$ . This term shifts the total energy by a constant and leaves the Heisenberg equations unchanged. An overall rescaling of the spectral parameter amounts to a corresponding normalization of the Hamiltonian. These normalization freedoms are fixed together with each explicit  $R$ -matrix.

To associate the  $R$ -matrix with a particular physical site, one copy of its second tensor factor is identified with the local Hilbert space  $\mathcal{H}_n$ . The resulting *quantum lattice Lax operator* is

$$L_{a,n}(u) := R_{a,n}(u) \in \text{End}(\mathcal{V}_a \otimes \mathcal{H}_n). \quad (2.12)$$

For a fixed physical site  $n$ , the Yang–Baxter equation implies the local exchange relation

$$R_{ab}(u-v) L_{a,n}(u) L_{b,n}(v) = L_{b,n}(v) L_{a,n}(u) R_{ab}(u-v), \quad (2.13)$$

acting on  $\mathcal{V}_a \otimes \mathcal{V}_b \otimes \mathcal{H}_n$ .

The ordered product of lattice Lax operators around the periodic chain defines the *quantum monodromy matrix*

$$T_a(u) := L_{a,N}(u) L_{a,N-1}(u) \cdots L_{a,1}(u), \quad (2.14)$$

and tracing over the auxiliary space gives the *transfer matrix*

$$\tau(u) := \text{tr}_a T_a(u). \quad (2.15)$$

Equation (2.13) implies

$$[\tau(u), \tau(v)] = 0, \quad (2.16)$$

so an expansion of  $\tau(u)$  generates mutually commuting quantum charges [15]. At the regular point, the logarithmic derivative of the transfer matrix reproduces the nearest-neighbour Hamiltonian (2.5), up to the scalar and normalization freedoms described after (2.11).

We next need the local equation governing the time evolution of  $L_{a,n}(u)$ . Differentiating the Yang–Baxter relation at its regular point and using (2.11) gives the *Sutherland relation* [6]

$$\begin{aligned} [h_{n-1,n}, L_{a,n}(u)L_{a,n-1}(u)] \\ = L_{a,n}(u)\partial_u L_{a,n-1}(u) - (\partial_u L_{a,n}(u))L_{a,n-1}(u). \end{aligned} \quad (2.17)$$

Here  $h_{n-1,n}$  denotes the embedding, defined above, of the same two-site Hamiltonian  $h$  on the neighbouring physical sites  $n-1$  and  $n$ .

Let  $t_{\text{lat}}$  denote the quantum lattice time. For an operator  $\mathcal{O}$  with no explicit time dependence we use the Heisenberg equation

$$\partial_{t_{\text{lat}}} \mathcal{O} = i[H^{\text{lat}}, \mathcal{O}]. \quad (2.18)$$

Equation (2.18) defines the algebraic time evolution used throughout the lattice Lax calculation. For a non-Hermitian Hamiltonian, normalized expectation values of right Schrödinger states in the standard inner product define a separate evolution problem. The continuum comparison below concerns the algebraic evolution in Eq. (2.18) and its coherent-state lower symbols. Because  $L_{a,n}$  acts only on the physical site  $n$ , only the two Hamiltonian terms  $h_{n-1,n}$  and  $h_{n,n+1}$  contribute to its commutator with  $H^{\text{lat}}$ . The Sutherland relation organizes these two contributions into a discrete zero-curvature equation.

For spectral parameters at which  $L_{a,n}(u)$  is invertible, define the *finite-lattice time Lax operator*

$$\begin{aligned} A_{a,n}(u) &= i h_{n-1,n} - i L_{a,n}(u)^{-1} h_{n-1,n} L_{a,n}(u) \\ &\quad - i L_{a,n}(u)^{-1} \partial_u L_{a,n}(u). \end{aligned} \quad (2.19)$$

The physical part of  $A_{a,n}$  acts on the neighbouring sites  $n-1$  and  $n$ , while its matrix indices belong to the auxiliary space  $\mathcal{V}_a$ . Equations (2.17) and (2.18) then give

$$\partial_{t_{\text{lat}}} L_{a,n}(u) = A_{a,n+1}(u)L_{a,n}(u) - L_{a,n}(u)A_{a,n}(u). \quad (2.20)$$

This is the lattice analogue of a zero-curvature equation. The time Lax operator has a scalar ambiguity: adding to every  $A_{a,n}$  the same scalar function of  $u$  times the identity on the full tensor product leaves (2.20) unchanged. This freedom permits convenient normalizations.

The continuum limit of (2.20) depends on the physical support of the operators in its two products. They contain operators that share a physical lattice site:  $A_{a,n+1}$  and  $L_{a,n}$  both act on  $\mathcal{H}_n$ , while  $L_{a,n}$  and  $A_{a,n}$  also overlap on  $\mathcal{H}_n$ . Consequently, coherent-state expectation values of these products retain correlations associated with the shared physical site.

## 2.2 Coherent states and long-wavelength limit

We next describe the common kinematical ingredients used to pass from the spin chain to a smooth classical spin field. We use standard  $SU(2)$  spin coherent states [11]; their use in long-wavelength limits of integrable spin chains is reviewed, for example, in [3]. The same construction is used in the direct continuum analysis of the Class 5 and Class 6 chains [7].

Let  $\sigma^1, \sigma^2, \sigma^3$  denote the Pauli matrices on the reference local Hilbert space  $\mathcal{H}_{\text{loc}} \simeq \mathbb{C}^2$ , and let  $|\uparrow\rangle$  and  $|\downarrow\rangle$  be the eigenvectors of  $\sigma^3$  with eigenvalues  $+1$  and  $-1$ , respectively. The spin- $\frac{1}{2}$  operators are normalized as

$$s^i := \frac{1}{2}\sigma^i, \quad i = 1, 2, 3. \quad (2.21)$$

Throughout, repeated Cartesian indices  $i, j, k \in \{1, 2, 3\}$  are summed. At the  $n$ th lattice site, a normalized spin coherent state is parametrized by polar and azimuthal angles  $\theta_n$  and  $\phi_n$  as

$$|\Omega_n\rangle = \cos \frac{\theta_n}{2} |\uparrow\rangle + e^{i\phi_n} \sin \frac{\theta_n}{2} |\downarrow\rangle. \quad (2.22)$$

This phase convention fixes the associated unit vector. We define the *classical spin vector at site  $n$*  by

$$\mathbf{S}_n := \langle \Omega_n | \boldsymbol{\sigma} | \Omega_n \rangle = \begin{pmatrix} \sin \theta_n \cos \phi_n \\ \sin \theta_n \sin \phi_n \\ \cos \theta_n \end{pmatrix}, \quad \boldsymbol{\sigma} := (\sigma^1, \sigma^2, \sigma^3)^\top. \quad (2.23)$$

It follows directly that

$$\mathbf{S}_n^\top \mathbf{S}_n = 1, \quad \langle \Omega_n | s^i | \Omega_n \rangle = \frac{1}{2} S_n^i. \quad (2.24)$$

Thus the normalization of the classical spin field is fixed independently of the model-dependent normalization of the Lax matrix.

For a configuration of the full chain we use the normalized product coherent state

$$|\Omega\rangle := |\Omega_1\rangle \otimes \cdots \otimes |\Omega_N\rangle \in \mathcal{H}^{(N)}. \quad (2.25)$$

The coherent-state labels are allowed to depend on time. We denote by  $t$  the time coordinate of the continuum field theory. Its relation to the quantum lattice time  $t_{\text{lat}}$  introduced in (2.18) is part of the continuum scaling.

Let  $\epsilon > 0$  denote the *lattice spacing*. We assign the spatial coordinate

$$x_n := (n-1)\epsilon, \quad n = 1, \dots, N, \quad (2.26)$$

and take the long-wavelength limit

$$\epsilon \rightarrow 0, \quad N \rightarrow \infty, \quad \ell := N\epsilon \text{ fixed}, \quad (2.27)$$

where  $\ell$  is the physical circumference of the continuum spatial circle. The long-wavelength assumption is that the coherent-state labels vary smoothly from one lattice site to the next. Equivalently, there is a smooth unit spin field  $\mathbf{S}(x, t)$  such that

$$\mathbf{S}_n(t) = \mathbf{S}(x_n, t), \quad \mathbf{S}(x + \ell, t) = \mathbf{S}(x, t). \quad (2.28)$$

For neighbouring sites this gives the Taylor expansion

$$\mathbf{S}_{n+1} = \mathbf{S} + \epsilon \partial_x \mathbf{S} + \frac{\epsilon^2}{2} \partial_x^2 \mathbf{S} + \mathcal{O}(\epsilon^3), \quad (2.29)$$

where the fields on the right-hand side are evaluated at  $(x_n, t)$ . Differentiating the unit-spin constraint  $\mathbf{S}^\top \mathbf{S} = 1$  gives the kinematical identities

$$\mathbf{S}^\top \partial_x \mathbf{S} = 0, \quad \mathbf{S}^\top \partial_x^2 \mathbf{S} = -(\partial_x \mathbf{S})^\top (\partial_x \mathbf{S}). \quad (2.30)$$

For any smooth scalar density  $f(x)$ , the corresponding lattice sum obeys the usual Riemann-sum relation

$$\epsilon \sum_{n=1}^N f(x_n) \longrightarrow \int_0^\ell dx f(x). \quad (2.31)$$

The continuum limit also rescales the Hamiltonian, time coordinate, and deformation parameters. The common time scaling is fixed in Section 2.3; the coupling scalings are determined in the two model sections.

We now define the operation that maps local quantum operators to classical quantities. Let  $X_a$  be an operator acting on the auxiliary space  $\mathcal{V}_a$  and on any finite set of physical lattice sites. Its *coherent-state lower symbol*, denoted by the superscript  $\downarrow$  [16], is the auxiliary-space operator

$$X_a^\downarrow := (\mathbf{1}_{\mathcal{V}_a} \otimes \langle \Omega |) X_a (\mathbf{1}_{\mathcal{V}_a} \otimes | \Omega \rangle). \quad (2.32)$$

Only the physical sites on which  $X_a$  acts contribute to the matrix element; the remaining factors in (2.25) cancel by normalization. If  $X$  has no auxiliary-space indices,  $X^\downarrow$  is a scalar function of the coherent-state labels. In particular, the lower symbol of the lattice Lax operator is

$$L_{a,n}^\downarrow(u) = \langle \Omega_n | L_{a,n}(u) | \Omega_n \rangle, \quad (2.33)$$

which is a  $2 \times 2$  matrix acting on the auxiliary space.

After evaluating these symbols on real spin vectors, we extend their algebraic expressions to complex spin components satisfying  $\mathbf{S}^\top \mathbf{S} = 1$ . Here  $\top$  denotes the transpose. Scalar and vector products are bilinear on this complexified spin space.

The product of two Pauli matrices on the same site satisfies  $\sigma^i \sigma^j = \delta_{ij} \mathbf{1}_2 + i \varepsilon_{ijk} \sigma^k$ , where  $\delta_{ij}$  is the Kronecker delta and  $\varepsilon_{ijk}$  is completely antisymmetric with  $\varepsilon_{123} = 1$ . Thus

$$(\sigma^i \sigma^j)^\downarrow - (\sigma^i)^\downarrow (\sigma^j)^\downarrow = \delta_{ij} - S^i S^j + i \varepsilon_{ijk} S^k. \quad (2.34)$$

Together with factorization between distinct physical sites, this identity evaluates all overlapping operator products used below.

The continuum spatial Lax matrix is extracted after the model-dependent normalization and spectral scaling have been fixed. Let  $\lambda$  denote the *continuum Lax spectral parameter*, to distinguish it from the quantum spectral parameter  $u$  defined in Section 2.1. For each model we choose a nonzero scalar normalization  $c_L(\epsilon, \lambda)$  and a relation  $u = u(\epsilon, \lambda)$  such that the normalized lattice operator

$$\hat{L}_{a,n}(\epsilon, \lambda) := c_L(\epsilon, \lambda) L_{a,n}(u(\epsilon, \lambda)) \quad (2.35)$$

has a regular long-wavelength expansion. When its lower symbol takes the form

$$\hat{L}_{a,n}^\downarrow(\epsilon, \lambda) = \mathbf{1}_{\mathcal{V}_a} + \epsilon U(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad (2.36)$$

we call the coefficient  $U(x, t; \lambda)$  the *continuum spatial Lax matrix*. The choices of  $c_L$ , of the scaling  $u(\epsilon, \lambda)$ , and of any deformation parameters are part of the model-specific continuum limit; they are specified separately for the two models.

The discrete zero-curvature equation (2.20) also contains products of operators with overlapping physical support. We next evaluate their lower symbols and determine their contribution to the continuum zero-curvature equation.

### 2.3 From the lattice Lax equation to the continuum zero-curvature equation

We now apply the lower-symbol operation to the finite-lattice equation (2.20). At this stage the lower symbol is taken at a fixed point of the coherent-state phase space. Applying it to the operator identity gives the exact relation

$$\begin{aligned} (\partial_{t_{\text{lat}}} L_{a,n}(u))^\downarrow &= (A_{a,n+1}(u) L_{a,n}(u))^\downarrow \\ &\quad - (L_{a,n}(u) A_{a,n}(u))^\downarrow. \end{aligned} \quad (2.37)$$

For each model, we derive the continuum Hamiltonian flow independently and use it to identify the continuum limit of the left-hand side with the time derivative of the classical spatial Lax matrix. The factors in each product on the right-hand side act on a common physical site. The lower symbol of each product contains a contribution in addition to the product of the separate lower symbols. We denote these contributions by the auxiliary-space matrices

$$\mathcal{D}_{a,n}^{(+)}(u) := (A_{a,n+1}(u) L_{a,n}(u))^\downarrow - A_{a,n+1}^\downarrow(u) L_{a,n}^\downarrow(u), \quad (2.38)$$

$$\mathcal{D}_{a,n}^{(-)}(u) := (L_{a,n}(u) A_{a,n}(u))^\downarrow - L_{a,n}^\downarrow(u) A_{a,n}^\downarrow(u). \quad (2.39)$$

These matrices measure the failure of the lower symbol to preserve the operator product at the overlapping physical site. This nonmultiplicativity is standard in coherent-state symbol calculus and is commonly encoded by a Berezin star product [12].

With the definitions (2.38) and (2.39), the lower-symbol form of the lattice Lax equation is the exact identity

$$\begin{aligned} (\partial_{t_{\text{lat}}} L_{a,n}(u))^\downarrow &= A_{a,n+1}^\downarrow(u) L_{a,n}^\downarrow(u) - L_{a,n}^\downarrow(u) A_{a,n}^\downarrow(u) \\ &\quad + \mathcal{D}_{a,n}^{(+)}(u) - \mathcal{D}_{a,n}^{(-)}(u). \end{aligned} \quad (2.40)$$

For operators with disjoint physical support, the product coherent state (2.25) makes the corresponding difference vanish. The difference terms in (2.40) can survive because of the overlap described at the end of Section 2.1.

We next state the common continuum scaling used by the two models studied in this paper. The model-dependent spectral scaling and scalar normalization of the spatial Lax operator have already been represented abstractly by (2.35). We also use the freedom described below (2.20) to shift the finite-lattice time Lax operator by a scalar function times the identity. We denote the resulting operator by  $\hat{A}_{a,n}$ .

For both models, the continuum time coordinate  $t$  is related to the quantum lattice time  $t_{\text{lat}}$  by

$$t = \epsilon^2 t_{\text{lat}}. \quad (2.41)$$

Equation (2.41) fixes the normalization of the continuum time used in this paper. Since the scalar normalization of  $\hat{L}_{a,n}$  and the scalar shift defining  $\hat{A}_{a,n}$  are independent of  $t_{\text{lat}}$  and of the lattice site, the normalized operators obey the same discrete zero-curvature equation as  $L_{a,n}$  and  $A_{a,n}$ . After the model-dependent spectral scaling has been imposed, the lower symbols of the normalized spatial and time Lax operators have the form

$$\hat{L}_{a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad (2.42)$$

$$\hat{A}_{a,n}^\downarrow = \epsilon^2 V^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3). \quad (2.43)$$

Here  $V^\downarrow(x, t; \lambda)$  denotes the continuum matrix obtained directly from the lower symbol of the finite-lattice time Lax operator. The scalings (2.41)–(2.43) are checked for each explicit model.

The part of (2.40) containing only products of lower symbols now has a standard continuum limit. Using  $x_{n+1} = x_n + \epsilon$  together with (2.42) and (2.43), one finds

$$\begin{aligned} \hat{A}_{a,n+1}^\downarrow \hat{L}_{a,n}^\downarrow - \hat{L}_{a,n}^\downarrow \hat{A}_{a,n}^\downarrow \\ = \epsilon^3 \left( \partial_x V^\downarrow + [V^\downarrow, U] \right)_{(x_n, t; \lambda)} + \mathcal{O}(\epsilon^4). \end{aligned} \quad (2.44)$$

The part containing products of lower symbols starts at order  $\epsilon^3$ . The coefficient of the lower symbol of the Heisenberg derivative at the same order is determined from the continuum Hamiltonian and its equations of motion for each model. This dynamical step is separate from the algebraic treatment of the operator products.

For the normalized operators, define the corresponding differences by

$$\hat{\mathcal{D}}_{a,n}^{(+)} := (\hat{A}_{a,n+1} \hat{L}_{a,n})^\downarrow - \hat{A}_{a,n+1}^\downarrow \hat{L}_{a,n}^\downarrow, \quad (2.45)$$

$$\hat{\mathcal{D}}_{a,n}^{(-)} := (\hat{L}_{a,n} \hat{A}_{a,n})^\downarrow - \hat{L}_{a,n}^\downarrow \hat{A}_{a,n}^\downarrow. \quad (2.46)$$

Their continuum contribution, when the limit exists, is

$$\mathcal{C}(x, t; \lambda) := \lim_{\epsilon \rightarrow 0} \frac{\hat{\mathcal{D}}_{a,n}^{(+)} - \hat{\mathcal{D}}_{a,n}^{(-)}}{\epsilon^3}, \quad x = x_n. \quad (2.47)$$

The evaluation of this limit requires both the operator expansion and the Taylor expansion of the spin arguments. Let  $m \in \{5, 6\}$  label the model, and write  $\mathcal{C}_m$  for its continuum contribution. For independent spin vectors  $\mathbf{S}_L$  and  $\mathbf{S}_R$  on the left and right physical sites, define the auxiliary-space matrix functions  $\mathcal{D}_{m,2}^{(\pm)}(\mathbf{S}_L, \mathbf{S}_R)$  and  $\mathcal{D}_{m,3}^{(\pm)}(\mathbf{S}_L, \mathbf{S}_R)$  as the coefficients

$$\widehat{\mathcal{D}}_{m;a,n}^{(+)} = \epsilon^2 \mathcal{D}_{m,2}^{(+)}(\mathbf{S}_n, \mathbf{S}_{n+1}) + \epsilon^3 \mathcal{D}_{m,3}^{(+)}(\mathbf{S}_n, \mathbf{S}_{n+1}) + \mathcal{O}(\epsilon^4), \quad (2.48)$$

$$\widehat{\mathcal{D}}_{m;a,n}^{(-)} = \epsilon^2 \mathcal{D}_{m,2}^{(-)}(\mathbf{S}_{n-1}, \mathbf{S}_n) + \epsilon^3 \mathcal{D}_{m,3}^{(-)}(\mathbf{S}_{n-1}, \mathbf{S}_n) + \mathcal{O}(\epsilon^4). \quad (2.49)$$

The dependence on the spectral parameter and continuum coupling is suppressed. The final subscript denotes the power of  $\epsilon$  in the operator-product difference before expanding the spin arguments. For both models, the quadratic coefficients agree at coincident spins,  $\mathcal{D}_{m,2}^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_{m,2}^{(-)}(\mathbf{S}, \mathbf{S})$ , as verified in the model calculations below. Taylor expansion at  $x = x_n$  then gives

$$\begin{aligned} \mathcal{C}_m = \partial_x S^i \left( \frac{\partial \mathcal{D}_{m,2}^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_{m,2}^{(-)}}{\partial S_L^i} \right) \Big|_{\mathbf{S}_L = \mathbf{S}_R = \mathbf{S}} \\ + \mathcal{D}_{m,3}^{(+)}(\mathbf{S}, \mathbf{S}) - \mathcal{D}_{m,3}^{(-)}(\mathbf{S}, \mathbf{S}). \end{aligned} \quad (2.50)$$

The Cartesian derivatives act on the polynomial lower symbols before imposing the unit-spin constraints and before setting the two arguments equal. Both derivatives have a plus sign: the shift  $\mathbf{S}_{n-1} = \mathbf{S} - \epsilon \partial_x \mathbf{S} + \dots$  occurs in the term subtracted in Eq. (2.47). Equation (2.50) separates the contribution from the spatial variation of the quadratic coefficients from the cubic coefficients at coincident spins.

Once the model-specific Hamiltonian check identifies the continuum limit of the lower symbol of the Heisenberg derivative on the left-hand side of (2.40) with  $\partial_t U$ , the order- $\epsilon^3$  equation becomes

$$\partial_t U = \partial_x V^\downarrow + [V^\downarrow, U] + \mathcal{C}. \quad (2.51)$$

The continuum zero-curvature equation is recovered if the matrix  $\mathcal{C}$  can be written in the local form

$$\mathcal{C} = \partial_x \Delta V + [\Delta V, U] \quad (2.52)$$

for some auxiliary-space matrix  $\Delta V(x, t; \lambda)$ . In that case the continuum time Lax matrix

$$V := V^\downarrow + \Delta V \quad (2.53)$$

satisfies

$$\partial_t U - \partial_x V + [U, V] = 0. \quad (2.54)$$

Equations (2.47)–(2.54) formulate the continuum problem. The matrix  $\Delta V$  is determined from the finite-lattice operator products for each model.

## 2.4 Shared-site correction and the Sutherland relation

For the scaling used below, the continuum source  $\mathcal{C}$  can be expressed in terms of the first coefficient of the normalized spatial Lax operator and the corresponding order of the Sutherland relation. Let  $P$  denote the permutation operator on two physical spin- $\frac{1}{2}$  spaces, and assume

$$\begin{aligned} \widehat{L} &= \mathbf{1} + \epsilon X + \epsilon^2 Y + \mathcal{O}(\epsilon^3), \\ h &= 2P + \epsilon h^{(1)} + \mathcal{O}(\epsilon^2), \\ \partial_u \widehat{L} &= \epsilon^2 Z + \mathcal{O}(\epsilon^3). \end{aligned} \quad (2.55)$$

Here  $X, Y, Z$  act on the auxiliary space and one physical site, while  $h^{(1)}$  acts on two neighbouring physical sites. The labels 1, 2 below specify the physical site on which a one-site operator acts;

the auxiliary space is shared and the order of auxiliary-space matrix products is retained. For the scalings of Sections 3 and 4,  $Z$  is the coefficient obtained by differentiating with respect to the quantum spectral parameter at fixed lattice coupling.

Define the order- $\epsilon^2$  Sutherland residual by

$$\mathcal{B} := [2P, X_2 X_1 + Y_2 + Y_1] + [h^{(1)}, X_2 + X_1] - Z_1 + Z_2. \quad (2.56)$$

For a lattice Lax operator satisfying the Sutherland relation at this order,  $\mathcal{B} = 0$ . We also use the symmetric two-spin projector  $P_+ := (\mathbf{1} + P)/2$  and impose

$$P_+ h^{(1)} P_+ = c P_+ \quad (2.57)$$

with a field-independent scalar  $c$ .

In Class 5 and Class 6, Eq. (2.57) follows from  $\mathcal{B} = 0$ . Expand

$$X = \mathbf{X}_0 \otimes \mathbf{1}_2 + \mathbf{X}_i \otimes \sigma^i. \quad (2.58)$$

If the three auxiliary-space matrices  $\mathbf{X}_i$  are linearly independent, the part of  $\mathcal{B} = 0$  that is even under exchange of the two physical sites implies

$$[h_+^{(1)}, \sigma_1^i + \sigma_2^i] = 0, \quad h_+^{(1)} := \frac{1}{2} (h^{(1)} + P h^{(1)} P), \quad i = 1, 2, 3. \quad (2.59)$$

The tensor product of two spin- $\frac{1}{2}$  representations contains one spin triplet and one spin singlet. An operator commuting with the three total-spin generators is therefore scalar on each sector. The exchange-odd part of  $h^{(1)}$  has vanishing triplet-to-triplet matrix elements, which gives Eq. (2.57). For both Class 5 and Class 6 the spin dependence has rank three for  $\lambda \neq 0$ ; one nonzero minor is

$$\det \frac{\partial(U_{12}, U_{21}, U_{11})}{\partial(S^1, S^2, S^3)} = \frac{i}{32\lambda^3}. \quad (2.60)$$

For a rank-deficient  $X$ , Eq. (2.57) need not follow from the Sutherland relation, so it remains an explicit assumption in the general formula.

At a fixed continuum spin  $\mathbf{S}$ , define

$$U := X^\downarrow, \quad \Xi := (X^2)^\downarrow - U^2, \quad (2.61)$$

where the operator square is formed before taking the lower symbol. The quantity  $\Xi$  is the finite-spin symbol correction associated with this operator product. Star-product formulations provide a general finite-spin symbol calculus for such products [13]. The local lattice time equation fixes how this correction enters the long-wavelength expansion. Expanding the finite-lattice time operator gives

$$\begin{aligned} A^{(1)} &= -i[2P, X_2], \\ A^{(2)} &= iX_2[2P, X_2] - i[2P, Y_2] - i[h^{(1)}, X_2] - iZ_2. \end{aligned} \quad (2.62)$$

We suppress the model label  $m$  on the coefficient functions  $\mathcal{D}_{m,2}^{(\pm)}$  and  $\mathcal{D}_{m,3}^{(\pm)}$  introduced in Eqs. (2.48) and (2.49). We also write  $\langle Q \rangle$  for the lower symbol of a two-physical-site operator  $Q$  evaluated with the same spin  $\mathbf{S}$  on both sites, leaving its auxiliary-space indices uncontracted. The analytic derivation of the next three identities is given in Appendix A.

At coincident spins, the order- $\epsilon^2$  coefficients satisfy

$$\mathcal{D}_2^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_2^{(-)}(\mathbf{S}, \mathbf{S}) = 2i\Xi. \quad (2.63)$$

Their first spatial Taylor coefficients give

$$\partial_x S^i \left( \frac{\partial \mathcal{D}_2^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_2^{(-)}}{\partial S_L^i} \right) \Big|_{\mathbf{S}_L = \mathbf{S}_R = \mathbf{S}} = 2i\partial_x \Xi. \quad (2.64)$$

The coincident order- $\epsilon^3$  coefficients obey

$$\mathcal{D}_3^{(+)}(\mathcal{S}, \mathcal{S}) - \mathcal{D}_3^{(-)}(\mathcal{S}, \mathcal{S}) = 2i[\Xi, U] - i\langle X_1 \mathcal{B} \rangle. \quad (2.65)$$

The explicit  $Y$  terms cancel after the contributions in Eq. (2.62) are combined. Using Eq. (2.50),

$$\mathcal{C} = 2i(\partial_x \Xi + [\Xi, U]) - i\langle X_1 \mathcal{B} \rangle. \quad (2.66)$$

For  $\mathcal{B} = 0$ ,

$$\mathcal{C} = 2i(\partial_x \Xi + [\Xi, U]), \quad (2.67)$$

which is absorbed by

$$\Delta V = 2i\Xi + f(\lambda)\mathbf{1}, \quad (2.68)$$

where  $f(\lambda)$  is independent of the spin and spacetime coordinates.

For spin  $\frac{1}{2}$ , Eq. (2.58) gives  $U = X_0 + X_i S^i$  and

$$\Xi = X_i X_j (\delta_{ij} - S^i S^j + i\varepsilon_{ijk} S^k). \quad (2.69)$$

In Class 6, Eq. (4.16) gives an equivalent representation of  $\Xi$  in terms of  $L_6^{(2)}$ .

### 3 Class 5 model

For Class 5, the shared-site source is a total spatial derivative. We derive the continuum time matrix from the lattice operator, compare its zero-curvature equation with the Hamiltonian flow, and then compare the same pair with classical constructions.

#### 3.1 Quantum Class 5 model and spatial Lax matrix

The name ‘‘Class 5’’ refers to one of the regular  $4 \times 4$  difference-form solutions of the Yang–Baxter equation found in the classification of nearest-neighbour integrable spin- $\frac{1}{2}$  chains in Ref. [6]. The long-wavelength limit of this chain is the deformed Landau–Lifshitz model studied in Ref. [7]. We use a one-parameter representative of the quantum chain for the quantum-to-classical Lax calculation. This representative is related to the general Class 5  $R$ -matrix by the spectral normalization and local basis transformation described in Ref. [7].

Let  $\kappa_5 \in \mathbb{C}$  denote the *Class 5 lattice deformation parameter*. It corresponds to the parameter denoted by  $a$  in Ref. [7]. In the ordered two-spin basis

$$\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}, \quad (3.1)$$

the two-site Hamiltonian operator defining the Class 5 chain in our normalization is

$$h_5(\kappa_5) = \begin{pmatrix} 2 & 2\kappa_5 & -2\kappa_5 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix}. \quad (3.2)$$

Following the embedding convention of Section 2.1, the copy of  $h_5$  on the bond joining sites  $n$  and  $n+1$  is denoted by  $h_{5;n,n+1}(\kappa_5)$ . The periodic quantum Hamiltonian is therefore

$$H_5^{\text{lat}}(\kappa_5) = \sum_{n=1}^N h_{5;n,n+1}(\kappa_5), \quad h_{5;N,N+1} \equiv h_{5;N,1}. \quad (3.3)$$

For  $\kappa_5 \neq 0$  this Hamiltonian is non-Hermitian in the standard inner product. The undeformed value  $\kappa_5 = 0$  reduces, up to the normalization fixed in Section 2.1, to the spin- $\frac{1}{2}$  XXX chain.

The same normalization is generated by the regular Class 5 quantum  $R$ -matrix

$$R_5(u; \kappa_5) = \begin{pmatrix} 1+2u & 2\kappa_5 u & -2\kappa_5 u & 0 \\ 0 & 2u & 1 & 0 \\ 0 & 1 & 2u & 0 \\ 0 & 0 & 0 & 1+2u \end{pmatrix}, \quad (3.4)$$

where  $u$  is the quantum spectral parameter introduced in Section 2.1. It obeys the Yang–Baxter equation and the regularity condition

$$R_5(0; \kappa_5) = P, \quad (3.5)$$

and a direct differentiation gives

$$h_5(\kappa_5) = P \left. \frac{\partial R_5(u; \kappa_5)}{\partial u} \right|_{u=0}, \quad (3.6)$$

in agreement with the general relation (2.11). Equations (3.2) and (3.4) fix the model and the normalizations used in this section.

We isolate the deformation in the two-site operator  $K_5$ , defined by

$$K_5 := \begin{pmatrix} 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \in \text{End}(\mathcal{H}_{\text{loc}} \otimes \mathcal{H}_{\text{loc}}). \quad (3.7)$$

With the permutation operator  $P$  defined in (2.9), the Class 5 data become

$$R_5(u; \kappa_5) = 2u\mathbf{1}_4 + P + 2\kappa_5 u K_5, \quad h_5(\kappa_5) = 2P + 2\kappa_5 K_5. \quad (3.8)$$

The operator  $K_5$  is nilpotent and satisfies

$$K_5^2 = 0, \quad PK_5 = K_5, \quad K_5 P = -K_5. \quad (3.9)$$

These identities simplify the finite-lattice time Lax operator.

We now specify the long-wavelength scaling. The lattice deformation is sent to zero together with the lattice spacing  $\epsilon$ , while the *Class 5 continuum deformation coupling*  $\alpha_5$  is held fixed:

$$\kappa_5 = \epsilon \alpha_5. \quad (3.10)$$

This is the scaling used in the direct continuum construction of Ref. [7]; it places the deformation and the ordinary Landau–Lifshitz terms at the same order in the continuum action. For the Lax operator we simultaneously relate the quantum spectral parameter  $u$  to the continuum Lax spectral parameter  $\lambda$  by

$$u = \frac{\lambda}{\epsilon}, \quad \lambda \neq 0, \quad (3.11)$$

and use the common time scaling  $t = \epsilon^2 t_{\text{lat}}$  introduced in (2.41).

For the fundamental spin- $\frac{1}{2}$  representation the lattice Lax operator is  $L_{5;a,n}(u) := R_{5;a,n}(u; \kappa_5)$ . We normalize it by its leading scalar part,

$$\hat{L}_{5;a,n}(\epsilon, \lambda) := \frac{1}{2u} L_{5;a,n}(u). \quad (3.12)$$

The quantum spectral derivatives are taken at fixed  $\kappa_5$ , before applying the scalings (3.10) and (3.11). The normalized operator has the exact expansion

$$\hat{L}_{5;a,n} = \mathbf{1}_{\mathcal{V}_a \otimes \mathcal{H}_n} + \epsilon L_{5;a,n}^{(1)}, \quad L_5^{(1)} = \frac{P}{2\lambda} + \alpha_5 K_5. \quad (3.13)$$

Here  $L_{5;a,n}^{(1)}$  denotes the copy of the fixed two-site operator  $L_5^{(1)}$  on  $\mathcal{V}_a \otimes \mathcal{H}_n$ .

To evaluate its lower symbol, define the classical transverse spin combinations

$$S^+(x, t) := S^1(x, t) + iS^2(x, t), \quad S^-(x, t) := S^1(x, t) - iS^2(x, t). \quad (3.14)$$

Throughout this section,  $S^\pm$  denotes these components of the classical unit spin field defined in (2.28). At the point  $x = x_n$ , the lower symbols of  $P_{a,n}$  and  $K_{5;a,n}$  are

$$P_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad (3.15)$$

$$K_{5;a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} S^+ & -(1 + S^3) \\ 0 & 0 \end{pmatrix}. \quad (3.16)$$

Substituting these expressions into (3.13) gives

$$\hat{L}_{5;a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U_5(x_n, t; \lambda), \quad (3.17)$$

where the *Class 5 continuum spatial Lax matrix* is

$$U_5(x, t; \lambda) = \frac{1}{4\lambda} \begin{pmatrix} 1 + S^3 + 2\alpha_5 \lambda S^+ & S^- - 2\alpha_5 \lambda (1 + S^3) \\ S^+ & 1 - S^3 \end{pmatrix}. \quad (3.18)$$

Ref. [7] constructs a classical spatial Lax matrix from the Class 5 quantum Lax operator. Equations (3.12)–(3.18) fix the relation used here at the fundamental spin- $\frac{1}{2}$  level: the normalized quantum Lax operator belongs to the same Class 5  $R/L$  family, and  $U_5$  is the coefficient of its spin- $\frac{1}{2}$  lower symbol in the long-wavelength scaling. Section 3.2 derives the time matrix associated with this continuum connection from the finite-lattice time operator. The finite-lattice limit retains the complete  $\mathfrak{gl}(2)$  matrix in Eq. (3.18), including its scalar part.

### 3.2 Time Lax matrix from the finite-lattice construction

We now take the long-wavelength limit of the finite-lattice time Lax operator. The starting point is the operator  $A_{a,n}(u)$  defined in Eq. (2.19). Adding the same scalar multiple of the identity to every time Lax operator leaves the lattice zero-curvature equation unchanged. We use the representative

$$\hat{A}_{5;a,n} := A_{5;a,n} + \frac{i}{u} \mathbf{1}, \quad (3.19)$$

where the identity acts on the auxiliary space and on the two physical sites  $n-1, n$  supporting  $A_{5;a,n}$ . In terms of the normalized spatial operator  $\hat{L}_{5;a,n}$  in Eq. (3.12),

$$\hat{A}_{5;a,n} = -i\hat{L}_{5;a,n}^{-1} \left( [h_{5;n-1,n}, \hat{L}_{5;a,n}] + \partial_u \hat{L}_{5;a,n} \right). \quad (3.20)$$

The derivative with respect to  $u$  is taken at fixed lattice deformation parameter  $\kappa_5$ . After imposing the continuum scalings (3.10) and (3.11),

$$\partial_u \hat{L}_{5;a,n} = -\frac{P_{a,n}}{2u^2} = -\epsilon^2 \frac{P_{a,n}}{2\lambda^2}. \quad (3.21)$$

The Hamiltonian on the neighbouring physical sites has the expansion

$$h_{5;n-1,n} = h_{n-1,n}^{(0)} + \epsilon h_{n-1,n}^{(1)}, \quad h^{(0)} := 2P, \quad h^{(1)} := 2\alpha_5 K_5. \quad (3.22)$$

Together with the exact relation  $\hat{L}_{5;a,n} = \mathbf{1} + \epsilon L_{5;a,n}^{(1)}$  in Eq. (3.13), Eq. (3.20) gives

$$\hat{A}_{5;a,n} = \epsilon A_{5;a,n}^{(1)} + \epsilon^2 A_{5;a,n}^{(2)} + \mathcal{O}(\epsilon^3), \quad (3.23)$$

with

$$A_{5;a,n}^{(1)} = -i[h_{n-1,n}^{(0)}, L_{5;a,n}^{(1)}], \quad (3.24)$$

$$A_{5;a,n}^{(2)} = iL_{5;a,n}^{(1)}[h_{n-1,n}^{(0)}, L_{5;a,n}^{(1)}] - i[h_{n-1,n}^{(1)}, L_{5;a,n}^{(1)}] + \frac{i}{2\lambda^2}P_{a,n}. \quad (3.25)$$

Both coefficients act on the tensor product of the auxiliary space and the two neighbouring physical sites.

We first evaluate the lower symbol of Eq. (3.23). Let  $\mathbf{S}_{n-1}$  and  $\mathbf{S}_n$  be the coherent-state spin vectors on the sites  $n-1$  and  $n$ . Direct contraction of Eq. (3.24) gives

$$(A_{5;a,n}^{(1)})^\downarrow = -2(\mathbf{S}_{n-1} \times \mathbf{S}_n)^i \frac{\partial U_5}{\partial S^i}. \quad (3.26)$$

The Cartesian derivatives act on Eq. (3.18), with  $S^\pm = S^1 \pm iS^2$ , before imposing the unit-spin constraint. They are independent of the spin because  $U_5$  is affine in its components. Using

$$\mathbf{S}_{n-1} = \mathbf{S}(x_n, t) - \epsilon \partial_x \mathbf{S}(x_n, t) + \mathcal{O}(\epsilon^2), \quad (3.27)$$

we have

$$\mathbf{S}_{n-1} \times \mathbf{S}_n = \epsilon \mathbf{S} \times \partial_x \mathbf{S} + \mathcal{O}(\epsilon^2). \quad (3.28)$$

The term  $A_{5;a,n}^{(2)}$  already carries the prefactor  $\epsilon^2$  in Eq. (3.23); its lower symbol is therefore needed only at coincident spin arguments at this order. Using the lower symbols in Eqs. (3.15) and (3.16), together with  $\mathbf{S}^\top \mathbf{S} = 1$ , gives

$$(A_{5;a,n}^{(2)})^\downarrow \Big|_{\mathbf{S}_{n-1}=\mathbf{S}_n=\mathbf{S}} = 4iU_5^2 - \frac{i}{2\lambda^2} \mathbf{1}_{\nu_a}. \quad (3.29)$$

Equations (3.26)–(3.29) then yield

$$\hat{A}_{5;a,n}^\downarrow = \epsilon^2 V_5^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3), \quad (3.30)$$

where

$$V_5^\downarrow = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_5}{\partial S^i} + 4iU_5^2 - \frac{i}{2\lambda^2} \mathbf{1}_{\nu_a}. \quad (3.31)$$

This is the continuum matrix obtained from the lower symbol of the finite-lattice time Lax operator itself.

The remaining contribution comes from the overlapping operator products in the discrete zero-curvature equation. For Class 5, denote the differences in Eqs. (2.45) and (2.46) by  $\hat{\mathcal{D}}_{5;a,n}^{(\pm)}$ . The quadratic coefficients in Eqs. (2.48) and (2.49) contain the products of  $A_5^{(1)}$  with  $L_5^{(1)}$ ; the cubic coefficients contain the products of  $A_5^{(2)}$  with  $L_5^{(1)}$ . Evaluating the shared-site Pauli products with Eq. (2.34) gives

$$\mathcal{D}_{5,2}^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_{5,2}^{(-)}(\mathbf{S}, \mathbf{S}) = 2i \left( \frac{\mathbf{1}_{\nu_a}}{4\lambda^2} - U_5^2 \right), \quad (3.32)$$

$$\partial_x S^i \left( \frac{\partial \mathcal{D}_{5,2}^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_{5,2}^{(-)}}{\partial S_L^i} \right) \Big|_{\mathbf{S}_L=\mathbf{S}_R=\mathbf{S}} = -2i\partial_x(U_5^2), \quad (3.33)$$

$$\mathcal{D}_{5,3}^{(+)}(\mathbf{S}, \mathbf{S}) - \mathcal{D}_{5,3}^{(-)}(\mathbf{S}, \mathbf{S}) = 0. \quad (3.34)$$

The first and third identities use the unit-spin constraint. The second also uses  $\mathbf{S}^\top \partial_x \mathbf{S} = 0$ . Thus the surviving contribution comes entirely from the first line of Eq. (2.50). In terms of the spatial operator,

$$\begin{aligned} & \hat{\mathcal{D}}_{5;a,n}^{(+)} - \hat{\mathcal{D}}_{5;a,n}^{(-)} \\ &= 2i\epsilon^3 \partial_x \left\{ \left[ (L_5^{(1)})^2 \right]^\downarrow - \left[ (L_5^{(1)})^\downarrow \right]^2 \right\} + \mathcal{O}(\epsilon^4). \end{aligned} \quad (3.35)$$

The algebra in Eq. (3.9) implies

$$(L_5^{(1)})^2 = \frac{1}{4\lambda^2} \mathbf{1}_4, \quad (3.36)$$

and Eq. (3.17) gives  $(L_5^{(1)})^\downarrow = U_5$ . Hence

$$\left[(L_5^{(1)})^2\right]^\downarrow - \left[(L_5^{(1)})^\downarrow\right]^2 = \frac{1}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a} - U_5^2, \quad (3.37)$$

and the continuum contribution defined in Eq. (2.47) is

$$\mathcal{C}_5 = -2i \partial_x (U_5^2). \quad (3.38)$$

Equation (3.38) is absorbed into the time component by choosing

$$\Delta V_5 := -2i U_5^2 + \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.39)$$

It obeys

$$[\Delta V_5, U_5] = 0, \quad \partial_x \Delta V_5 + [\Delta V_5, U_5] = \mathcal{C}_5. \quad (3.40)$$

The scalar term in Eq. (3.39) fixes the representative; an  $x$ -independent scalar function of  $\lambda$  leaves the curvature unchanged. Combining Eqs. (3.31) and (3.39) gives

$$V_5 = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_5}{\partial S^i} + 2i U_5^2 - \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a}. \quad (3.41)$$

We write  $S_x^a := \partial_x S^a$  for  $a \in \{+, -, 3\}$ . Substituting Eq. (3.18) and using the unit-spin constraint gives the matrix elements

$$(V_5)_{11} = \frac{i}{4\lambda^2} \left[ S^3 + \alpha_5 \lambda S^+ (1 + S^3) + 2\alpha_5^2 \lambda^2 (S^+)^2 \right. \\ \left. + \lambda (S^- S_x^+ - S^+ S_x^-) + 4\alpha_5 \lambda^2 (S^+ S_x^3 - S^3 S_x^+) \right], \quad (3.42)$$

$$(V_5)_{12} = \frac{i}{4\lambda^2} \left[ S^- - \alpha_5 \lambda (1 + S^3)^2 - 2\alpha_5^2 \lambda^2 S^+ (1 + S^3) \right. \\ \left. + 2\lambda (S^3 S_x^- - S^- S_x^3) + 2\alpha_5 \lambda^2 (S^+ S_x^- - S^- S_x^+) \right], \quad (3.43)$$

$$(V_5)_{21} = \frac{i}{4\lambda^2} \left[ S^+ + \alpha_5 \lambda (S^+)^2 + 2\lambda (S^+ S_x^3 - S^3 S_x^+) \right], \quad (3.44)$$

$$(V_5)_{22} = \frac{i}{4\lambda^2} \left[ -S^3 - \alpha_5 \lambda S^+ (1 + S^3) + \lambda (S^+ S_x^- - S^- S_x^+) \right]. \quad (3.45)$$

The difference between Eqs. (3.31) and (3.41) is the contribution required by the overlapping operator products in the finite-lattice zero-curvature equation.

### 3.3 Zero-curvature equation and the Landau–Lifshitz equations of motion

We now determine the continuum Hamiltonian flow independently of the time Lax matrix derived in Section 3.2. This fixes the dynamical identification left open in Section 2.3 and provides a direct test of the corrected matrix  $V_5$ .

It is convenient to introduce the constant antisymmetric matrix

$$M_5 := \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -i \\ 1 & i & 0 \end{pmatrix}, \quad M_5^T = -M_5. \quad (3.46)$$

With  $\mathbf{S}_n(t) = \mathbf{S}(x_n, t)$  and  $\kappa_5 = \epsilon \alpha_5$ , the lower symbol of the Class 5 bond Hamiltonian has the long-wavelength expansion

$$h_{5;n,n+1}^\downarrow = 2 + \epsilon^2 \mathcal{H}_5(x_n, t) + \mathcal{O}(\epsilon^3), \quad (3.47)$$

where the Hamiltonian density at  $(x, t)$  is

$$\mathcal{H}_5(x, t) = -\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \frac{\alpha_5}{2} \left( \mathbf{S}^\top M_5 \partial_x \mathbf{S} + \partial_x S^+ \right). \quad (3.48)$$

The last term is a total spatial derivative. With the periodic boundary condition in (2.28), the continuum Hamiltonian obtained from the lattice Hamiltonian is therefore

$$\begin{aligned} H_5^{\text{cl}} &:= \lim_{\epsilon \rightarrow 0} \frac{(H_5^{\text{lat}})^\downarrow - 2N}{\epsilon} \\ &= \int_0^\ell dx \left[ -\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S} + \frac{\alpha_5}{2} \mathbf{S}^\top M_5 \partial_x \mathbf{S} \right]. \end{aligned} \quad (3.49)$$

The signs and the overall normalization in (3.49) follow from the lattice Hamiltonian (3.3), the Heisenberg equation (2.18), and the continuum time convention  $t = \epsilon^2 t_{\text{lat}}$ .

For the unit spin field normalized in (2.28), we use the continuum Poisson bracket

$$\{S^i(x), S^j(y)\}_5 = 2 \varepsilon_{ijk} S^k(x) \delta(x - y), \quad \varepsilon_{123} = 1, \quad (3.50)$$

where  $\delta(x - y)$  is the Dirac delta function and the common time argument is suppressed. Varying the Cartesian spin components before imposing the unit-spin constraint, and integrating by parts with periodic variations, gives  $\delta H_5^{\text{cl}} / \delta \mathbf{S} = \partial_x^2 \mathbf{S} + \alpha_5 M_5 \partial_x \mathbf{S}$ . The corresponding Hamiltonian flow is

$$\partial_t S^i(x) = \{S^i(x), H_5^{\text{cl}}\}_5. \quad (3.51)$$

Using the bilinear cross product  $(\mathbf{A} \times \mathbf{B})^i = \varepsilon_{ijk} A^j B^k$ , the equations of motion take the compact form

$$\partial_t \mathbf{S} = -2 \mathbf{S} \times \left( \partial_x^2 \mathbf{S} + \alpha_5 M_5 \partial_x \mathbf{S} \right). \quad (3.52)$$

The right-hand side is tangent to the unit-spin constraint, so  $\partial_t(\mathbf{S}^\top \mathbf{S}) = 0$ .

For comparison with the matrix entries of the Lax pair, the same equations can be written in terms of the transverse components  $S^\pm$  defined in (3.14). Using  $\mathbf{S}^\top \partial_x \mathbf{S} = 0$ , one obtains

$$\partial_t S^+ = 2i \left( S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha_5 S^+ \partial_x S^+ \right), \quad (3.53)$$

$$\partial_t S^- = 2i \left( S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + \alpha_5 S^+ \partial_x S^- \right), \quad (3.54)$$

$$\partial_t S^3 = i \left( S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- + 2\alpha_5 S^+ \partial_x S^3 \right). \quad (3.55)$$

The same flow follows directly from the finite-lattice Heisenberg derivative. Let  $\boldsymbol{\sigma}_n = (\sigma_n^1, \sigma_n^2, \sigma_n^3)^\top$ , where  $\sigma_n^i$  is the Pauli matrix  $\sigma^i$  acting at site  $n$ . The two bonds adjacent to that site give

$$\begin{aligned} \frac{1}{\epsilon^2} (\partial_{t_{\text{lat}}} \boldsymbol{\sigma}_n)^\downarrow &= -2 \mathbf{S}_n \times \left[ \frac{\mathbf{S}_{n-1} + \mathbf{S}_{n+1} - 2 \mathbf{S}_n}{\epsilon^2} \right. \\ &\quad \left. + \frac{\alpha_5}{2\epsilon} M_5 (\mathbf{S}_{n+1} - \mathbf{S}_{n-1}) \right]. \end{aligned} \quad (3.56)$$

Its smooth-field limit is Eq. (3.52). Applying the same commutator to the normalized lattice Lax operator gives

$$(\partial_{t_{\text{lat}}} \widehat{L}_{5;a,n})^\downarrow = \epsilon^3 \partial_t U_5(x_n, t; \lambda) + \mathcal{O}(\epsilon^4), \quad (3.57)$$

with  $\partial_t \mathbf{S}$  given by (3.52). Equation (3.57) therefore supplies the dynamical identification required on the left-hand side of the continuum equation (2.51).

To establish the off-shell relation between the curvature and the equations of motion, we treat the time derivatives as independent variables and define the three equation-of-motion residuals

$$E_+^{(5)} := \partial_t S^+ - 2i \left( S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha_5 S^+ \partial_x S^+ \right), \quad (3.58)$$

$$E_-^{(5)} := \partial_t S^- - 2i \left( S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + \alpha_5 S^+ \partial_x S^- \right), \quad (3.59)$$

$$E_3^{(5)} := \partial_t S^3 - i \left( S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- + 2\alpha_5 S^+ \partial_x S^3 \right). \quad (3.60)$$

The curvature of the matrices  $U_5$  and  $V_5$  in (3.18) and (3.41) is

$$F_{tx}^{(5)} := \partial_t U_5 - \partial_x V_5 + [U_5, V_5]. \quad (3.61)$$

Using only the unit-spin constraint and its spatial derivatives, the curvature factorizes as

$$F_{tx}^{(5)} = \frac{1}{4\lambda} \begin{pmatrix} E_3^{(5)} + 2\alpha_5 \lambda E_+^{(5)} & E_-^{(5)} - 2\alpha_5 \lambda E_3^{(5)} \\ E_+^{(5)} & -E_3^{(5)} \end{pmatrix}. \quad (3.62)$$

For  $\lambda \neq 0$ , the residuals are recovered from the curvature entries by

$$\begin{aligned} E_+^{(5)} &= 4\lambda (F_{tx}^{(5)})_{21}, & E_3^{(5)} &= -4\lambda (F_{tx}^{(5)})_{22}, \\ E_-^{(5)} &= 4\lambda \left( (F_{tx}^{(5)})_{12} - 2\alpha_5 \lambda (F_{tx}^{(5)})_{22} \right). \end{aligned} \quad (3.63)$$

Hence the zero-curvature condition  $F_{tx}^{(5)} = 0$  is equivalent to the three continuum equations of motion (3.53)–(3.55).

To compare with the direct continuum construction of Ref. [7], denote its spin field, coupling, and time by  $\mathbf{S}_{\text{ref}}$ ,  $\alpha_{\text{ref}}$ , and  $t_{\text{ref}}$ . The convention dictionary is

$$\begin{aligned} t_{\text{ref}} &= 2t, & \alpha_{\text{ref}} &= -\alpha_5, \\ \mathbf{S}_{\text{ref}}(x, t_{\text{ref}}) &= \text{diag}(1, -1, 1) \mathbf{S}(x, t_{\text{ref}}/2). \end{aligned} \quad (3.64)$$

With these identifications, Eq. (3.52) takes the form of the Class 5 continuum equation used in Ref. [7]. The reflection of the second spin axis conjugates  $M_5$  by  $\text{diag}(1, -1, 1)$ , yielding the deformation matrix used there.

### 3.4 Comparison with the classical $r$ -matrix construction

The time Lax matrix in Eq. (3.41) was obtained by starting from the finite-lattice time operator. We now construct the time component independently from the classical  $r$ -matrix and monodromy of the continuum model.

To keep the two roles of the spectral parameter separate, let  $\zeta \in \mathbb{C}$  denote the *generating spectral parameter* used to expand the transfer matrix, while  $\lambda$  continues to denote the Lax spectral parameter of the final pair  $(U_5, V_5)$ . Define the rank-one coherent-state projector

$$\rho(x, t) := \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad \rho^2 = \rho, \quad \text{tr } \rho = 1, \quad (3.65)$$

and the deformation matrix in the auxiliary space

$$K_5^{\text{cl}}(x, t) := \frac{1}{2} \begin{pmatrix} S^+ & -(1 + S^3) \\ 0 & 0 \end{pmatrix}. \quad (3.66)$$

These are precisely the lower symbols in Eqs. (3.15) and (3.16). The spatial Lax matrix therefore admits the form

$$U_5(x, t; \lambda) = \frac{\rho(x, t)}{2\lambda} + \alpha_5 K_5^{\text{cl}}(x, t). \quad (3.67)$$

We suppress the common time argument in the following formulas and first fix the classical  $r$ -matrix in the conventions of this paper. Let  $P_{12}$  be the permutation operator on two copies of the two-dimensional auxiliary space. On either copy, define the projector  $P_\uparrow$  and the matrix unit  $E_{12}$  by

$$P_\uparrow := \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad E_{12} := \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad (3.68)$$

and the constant Jordanian tensor

$$C_{12} := P_\uparrow \otimes E_{12} - E_{12} \otimes P_\uparrow. \quad (3.69)$$

A direct calculation with the spin Poisson bracket (3.50) gives

$$r_{12}^{(5)}(\zeta, \lambda) = \frac{i}{2(\zeta - \lambda)} P_{12} + i\alpha_5 C_{12}, \quad (3.70)$$

for which

$$\{U_{5,1}(x; \zeta), U_{5,2}(y; \lambda)\}_5 = [r_{12}^{(5)}(\zeta, \lambda), U_{5,1}(x; \zeta) + U_{5,2}(y; \lambda)] \delta(x - y). \quad (3.71)$$

Here  $U_{5,1} := U_5 \otimes \mathbf{1}_2$  and  $U_{5,2} := \mathbf{1}_2 \otimes U_5$  act on the first and second auxiliary spaces, respectively. The rational part of Eq. (3.70) is the usual isotropic Landau–Lifshitz contribution, whereas the constant term contains the Class 5 deformation.

Ref. [7] gives a quadratic Poisson algebra for the classical Lax matrix obtained from its Class 5 quantum Lax operator. Equation (3.71) concerns the long-wavelength coefficient  $U_5$  defined by Eq. (3.17). Its bracket follows directly from the continuum spin Poisson bracket (3.50). The relation of the underlying quantum representative to  $U_5$  is fixed in Section 3.1, with the continuum normalization of Eqs. (3.10)–(3.12).

For the representative used here, Eq. (37) of Ref. [7] gives the classical  $r$ -matrix

$$r_{\text{ref}}(u; \kappa_5) = - \left( \frac{R_5(u; \kappa_5)}{2u} - \mathbf{1}_4 \right) = - \frac{P}{2u} - \kappa_5 K_5. \quad (3.72)$$

Since  $K_5 = C_{12}$  in the basis used above, the spectral and deformation scalings give

$$r_{12}^{(5)}(\zeta, \lambda) = - \frac{i}{\epsilon} r_{\text{ref}} \left( \frac{\zeta - \lambda}{\epsilon}; \epsilon \alpha_5 \right), \quad (3.73)$$

which reproduces Eq. (3.70).

Let  $\mathcal{P}$  denote path ordering with larger spatial coordinates to the left. The classical monodromy matrix is

$$T(x, y; \zeta) := \mathcal{P} \exp \left( \int_y^x U_5(s; \zeta) ds \right). \quad (3.74)$$

The associated transfer matrix on the periodic interval of length  $\ell$  is

$$\tau(\zeta) := \text{tr} T(\ell, 0; \zeta). \quad (3.75)$$

For a linear classical  $r$ -matrix algebra of the form (3.71), the standard monodromy construction [17, 4] gives the generating time matrix

$$\mathbb{V}_2(x; \zeta, \lambda) = \frac{1}{\tau(\zeta)} \text{tr}_1 \left[ T_1(\ell, x; \zeta) r_{12}^{(5)}(\zeta, \lambda) T_1(x, 0; \zeta) \right], \quad (3.76)$$

where the trace is taken over the first auxiliary space. For  $\tau(\zeta) \neq 0$  and  $\zeta \neq \lambda$ , it satisfies

$$\{\log \tau(\zeta), U_{5,2}(x; \lambda)\}_5 = \partial_x \mathbb{V}_2 + [\mathbb{V}_2, U_{5,2}(x; \lambda)]. \quad (3.77)$$

The pole of  $U_5(x; \zeta)$  at  $\zeta = 0$  has residue  $\rho/2$ , by Eq. (3.67), with eigenvalues  $1/2$  and  $0$ . To extract local charges, we use the formal small- $\zeta$  expansion associated with the first eigenvalue. In a sector where  $\text{Re}(1/\zeta) > 0$ , the corresponding monodromy eigenvalue has the dominant exponential factor  $\exp(\ell/(2\zeta))$ . Thus

$$\frac{T(x, 0; \zeta)T(\ell, x; \zeta)}{\tau(\zeta)} \sim \Pi(x; \zeta), \quad (3.78)$$

where  $\sim$  denotes the formal power-series expansion, with the exponentially small contribution of the other eigenvalue omitted. The normalized monodromy at finite  $\zeta$  has trace one; its local expansion is described by the rank-one projector  $\Pi$  satisfying

$$\Pi^2 = \Pi, \quad \text{tr} \Pi = 1, \quad \partial_x \Pi = [U_5(x; \zeta), \Pi], \quad (3.79)$$

with the branch

$$\Pi(x; \zeta) = \rho(x) + \zeta \Pi_1(x) + \mathcal{O}(\zeta^2). \quad (3.80)$$

The projector constraint at order  $\zeta$  and the transport equation at order  $\zeta^0$  give

$$\begin{aligned} \rho \Pi_1 + \Pi_1 \rho &= \Pi_1, \\ \partial_x \rho &= \frac{1}{2}[\rho, \Pi_1] + \alpha_5[K_5^{\text{cl}}, \rho]. \end{aligned} \quad (3.81)$$

These determine the coefficient uniquely as

$$\Pi_1 = 2[\rho, \partial_x \rho] - 2\alpha_5[\rho, [K_5^{\text{cl}}, \rho]]. \quad (3.82)$$

Using Eq. (3.80) in the generating matrix (3.76), the local series reads

$$\mathbb{V}_2(x; \zeta, \lambda) = \mathbb{V}^{(0)}(x; \lambda) + \zeta \mathbb{V}^{(1)}(x; \lambda) + \mathcal{O}(\zeta^2). \quad (3.83)$$

The coefficient of  $\zeta$  is

$$\begin{aligned} \mathbb{V}^{(1)} &= -\frac{i}{2\lambda} \left( \Pi_1 + \frac{\rho}{\lambda} \right) \\ &\quad + i\alpha_5 [\text{tr}(\Pi_1 P_\uparrow) E_{12} - \text{tr}(\Pi_1 E_{12}) P_\uparrow]. \end{aligned} \quad (3.84)$$

Equation (3.77) places the generating Hamiltonian in the first argument of the Poisson bracket. Antisymmetry of the bracket converts this to the evolution convention in Eq. (3.51). The coefficient of  $\zeta$  in  $\log \tau(\zeta)$  therefore has the time Lax matrix

$$V_{5,r}(x; \lambda) := -\mathbb{V}^{(1)}(x; \lambda). \quad (3.85)$$

Substituting Eq. (3.82) into Eq. (3.84) and using only the unit spin constraint gives

$$V_{5,r}(x; \lambda) - V_5(x; \lambda) = \frac{i}{4\lambda^2} \mathbf{1}_2. \quad (3.86)$$

The matrix on the right-hand side is independent of the spin field and of the spatial coordinate. Its spatial derivative vanishes and it commutes with  $U_5$ , so  $V_{5,r}$  and  $V_5$  generate the same zero-curvature equation. This is the scalar freedom described after Eq. (3.39). The comparison identifies the associated local charge as generating the  $H_5^{\text{cl}}$  spin flow. This identification concerns the Hamiltonian vector field on the unit-spin constraint; additive constants, Casimirs, and total derivatives with vanishing periodic integral leave it unchanged.

### 3.5 Comparison with the null-like warped $SL(2)$ Landau–Lifshitz model

The null-like warped  $SL(2)$  Landau–Lifshitz model of Ref. [2]<sup>1</sup> is defined in its Sec. 5.1 by a first-order Landau–Lifshitz action. Let  $n^0, n^1, n^2$  denote its spin variables and set

$$(n^0)^2 - (n^1)^2 - (n^2)^2 = 1, \quad n^\pm := \frac{n^0 \pm n^1}{\sqrt{2}}, \quad \eta := \text{diag}(1, -1, -1). \quad (3.87)$$

Writing the constants in that action as  $L_K$ ,  $\Lambda_K$  and  $C_K$ , define

$$A_K := \frac{4\pi L_K}{\sqrt{\Lambda_K}}. \quad (3.88)$$

The Hamiltonian density following from the action, with its overall factor  $L_K/2$  stripped off, is

$$\mathcal{H}_K = C_K (n^-)^2 - \frac{1}{A_K^2} \partial_x n^a \eta_{ab} \partial_x n^b. \quad (3.89)$$

Here and below  $a, b = 0, 1, 2$  are summed.

The position-dependent rotation used below appears in Ref. [7], Eq. (31), with the rotated Hamiltonian given in its Eq. (32). In the conventions of this paper the rotation is

$$\mathbf{S}(x, t) = \mathcal{R}(x) \tilde{\mathbf{S}}(x, t), \quad \mathcal{R}(x) := \exp\left(-\frac{\alpha_5 x}{2} M_5\right). \quad (3.90)$$

Since  $M_5^\top = -M_5$ , the matrix  $\mathcal{R}$  is complex orthogonal. Substitution into the Class 5 Hamiltonian density gives

$$\tilde{\mathcal{H}}_5 = -\frac{1}{2} \partial_x \tilde{\mathbf{S}}^\top \partial_x \tilde{\mathbf{S}} - \frac{\alpha_5^2}{8} \tilde{\mathbf{S}}^\top M_5^2 \tilde{\mathbf{S}}. \quad (3.91)$$

We use this rotated form to compare the Class 5 dynamics with the null-like warped model. Set

$$\begin{pmatrix} n^0(x, t_K) \\ n^1(x, t_K) \\ n^2(x, t_K) \end{pmatrix} = \text{diag}(1, -i, -i) \tilde{\mathbf{S}}(x, t_K/A_K^2), \quad C_K = \frac{\alpha_5^2}{2A_K^2}, \quad t_K = A_K^2 t. \quad (3.92)$$

The identities

$$\text{diag}(1, -i, -i)^\top \eta \text{diag}(1, -i, -i) = \mathbf{1}_3, \quad \tilde{\mathbf{S}}^\top M_5^2 \tilde{\mathbf{S}} = -2(n^-)^2 \quad (3.93)$$

then give

$$\tilde{\mathcal{H}}_5 = \frac{A_K^2}{2} \mathcal{H}_K. \quad (3.94)$$

The same field redefinition maps the Class 5 equation (3.52) to

$$\partial_t \tilde{\mathbf{S}} = -2\tilde{\mathbf{S}} \times \left( \partial_x^2 \tilde{\mathbf{S}} - \frac{\alpha_5^2}{4} M_5^2 \tilde{\mathbf{S}} \right), \quad (3.95)$$

and Eq. (3.92) converts its three components into the equations of motion of the null-like warped model in Ref. [2]. The comparison below is therefore between Lax representations of locally equivalent complexified equations of motion, with the Hamiltonian densities related by Eq. (3.94). No identification of the two real phase spaces or of their Poisson structures is assumed here.

For the Lax comparison, introduce

$$T^0 = \frac{i}{2} \sigma^2, \quad T^1 = \frac{1}{2} \sigma^1, \quad T^2 = \frac{1}{2} \sigma^3, \quad T^\pm = \frac{T^0 \pm T^1}{\sqrt{2}}, \quad B_K := -\frac{2}{A_K}, \quad (3.96)$$

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<sup>1</sup>Section and equation numbers follow the published version of Ref. [2].

and relate the spectral parameters by

$$z_K = 2iA_K\lambda. \quad (3.97)$$

In these conventions the Lax pair of Ref. [2], Eqs. (5.7) and (5.8), is

$$U_K = \frac{A_K}{z_K} \left[ - \left( n^+ - \frac{C_K z_K^2}{2} n^- \right) T^- + n^2 T^2 - n^- T^+ \right], \quad (3.98)$$

$$\begin{aligned} V_K = \frac{B_K}{z_K} \left[ - \left( n^2 \partial_x n^+ - n^+ \partial_x n^2 - \frac{C_K z_K^2}{2} (n^- \partial_x n^2 - n^2 \partial_x n^-) \right) T^- \right. \\ \left. + (n^- \partial_x n^+ - n^+ \partial_x n^-) T^2 - (n^- \partial_x n^2 - n^2 \partial_x n^-) T^+ \right] \\ + \frac{A_K B_K}{z_K^2} \left[ - \left( n^+ + \frac{C_K z_K^2}{2} n^- \right) T^- + n^2 T^2 - n^- T^+ \right]. \end{aligned} \quad (3.99)$$

For the auxiliary-space transformation, let

$$Q = \begin{pmatrix} 0 & i \\ 1 & 0 \end{pmatrix}, \quad G(x, \lambda) = \mathbf{1}_2 + \alpha_5 \left( \lambda - \frac{x}{2} \right) E_{12}, \quad \mathcal{G} := GQ^{-1}, \quad (3.100)$$

where  $E_{12}$  is the matrix unit with only its  $(1, 2)$  entry nonzero. Define

$$U_{5,0} := U_5 - \frac{1}{2} \text{tr } U_5 \mathbf{1}_2, \quad V_{5,0} := V_5 - \frac{1}{2} \text{tr } V_5 \mathbf{1}_2. \quad (3.101)$$

Then

$$U_{5,0} = \mathcal{G} U_K \mathcal{G}^{-1} + (\partial_x \mathcal{G}) \mathcal{G}^{-1}, \quad (3.102)$$

$$V_{5,0} = A_K^2 \mathcal{G} V_K \mathcal{G}^{-1} \quad (3.103)$$

on the unit-spin constraint. The spatial identity holds without the constraint; before imposing it, the difference in the time identity is proportional to  $\tilde{\mathbf{S}}^\top \tilde{\mathbf{S}} - 1$ .

The scalar parts are

$$a_U := \frac{1}{2} \text{tr } U_5, \quad a_V := \frac{1}{2} \text{tr } V_5, \quad (3.104)$$

and, on the unit-spin constraint,

$$\begin{aligned} a_U &= \frac{1}{4\lambda} + \frac{\alpha_5}{4} S^+, \\ a_V &= \frac{i\alpha_5}{2} (S^+ \partial_x S^3 - S^3 \partial_x S^+) + \frac{i\alpha_5^2}{4} (S^+)^2. \end{aligned} \quad (3.105)$$

Using the residual  $E_+^{(5)}$  defined in Eq. (3.58),

$$\partial_t a_U - \partial_x a_V = \frac{\alpha_5}{4} E_+^{(5)}. \quad (3.106)$$

The scalar sector is locally flat on the Class 5 equations of motion and can be removed on shell by a local scalar auxiliary transformation.

The equivalence above is local. For a periodic Class 5 spin with circumference  $\ell$ ,

$$\tilde{\mathbf{S}}(x + \ell, t) = \exp\left(\frac{\alpha_5 \ell}{2} M_5\right) \tilde{\mathbf{S}}(x, t), \quad G(x + \ell, \lambda) G(x, \lambda)^{-1} = \mathbf{1}_2 - \frac{\alpha_5 \ell}{2} E_{12}. \quad (3.107)$$

The field and gauge transformations are not generically periodic; periodic monodromies and global boundary sectors are therefore not identified by this map.

## 4 Class 6 model

We now apply the finite-lattice construction to the Class 6 chain, using the quantum-chain normalization of Appendix A of Ref. [7]. The spatial expansion contains a nonzero second-order operator coefficient. We determine its contribution to the time Lax matrix, including the commutator term in the lower-symbol correction, and then establish the off-shell equivalence of the corrected Lax equation and the Hamiltonian equations of motion.

### 4.1 Quantum Class 6 model and spatial Lax matrix

Let  $\kappa_6 \in \mathbb{C}$  denote the *Class 6 lattice deformation parameter*. We use the ordered two-spin basis  $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$  and fix the coefficient of the permutation interaction to be 2. In this normalization the local Hamiltonian is

$$h_6(\kappa_6) = \begin{pmatrix} 2 & \kappa_6 & \kappa_6 & 0 \\ 0 & 0 & 2 & -\kappa_6 \\ 0 & 2 & 0 & -\kappa_6 \\ 0 & 0 & 0 & 2 \end{pmatrix} = 2P + \kappa_6 K_6, \quad (4.1)$$

where  $P$  is the permutation operator in Eq. (2.9), and the two-site deformation operator  $K_6$  is defined by

$$K_6 := \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.2)$$

Its algebra with  $P$  is

$$PK_6 = K_6P = K_6, \quad K_6^3 = 0, \quad K_6^2 = \begin{pmatrix} 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (4.3)$$

For the periodic chain of  $N$  sites, the quantum Hamiltonian is

$$H_6^{\text{lat}}(\kappa_6) = \sum_{n=1}^N h_{6;n,n+1}(\kappa_6), \quad h_{6;N,N+1} \equiv h_{6;N,1}. \quad (4.4)$$

The Hamiltonian is non-Hermitian for  $\kappa_6 \neq 0$  in the standard inner product. At  $\kappa_6 = 0$  it reduces to the spin- $\frac{1}{2}$  XXX chain with the same exchange normalization as in Section 3.1.

Using the quantum spectral parameter  $u$  of Section 2.1, the regular  $R$ -matrix generating Eq. (4.1) is

$$R_6(u; \kappa_6) = 2u\mathbf{1}_4 + P + \kappa_6 u(1+2u)K_6 + \frac{\kappa_6^2 u^2 (1+2u)^2}{2} K_6^2 \quad (4.5)$$

$$= \begin{pmatrix} 1+2u & \kappa_6 u(1+2u) & \kappa_6 u(1+2u) & -\kappa_6^2 u^2 (1+2u)^2 \\ 0 & 2u & 1 & -\kappa_6 u(1+2u) \\ 0 & 1 & 2u & -\kappa_6 u(1+2u) \\ 0 & 0 & 0 & 1+2u \end{pmatrix}. \quad (4.6)$$

This is Eq. (A.1) of Ref. [7] with the exchange normalization fixed above. It satisfies the Yang–Baxter equation and

$$R_6(0; \kappa_6) = P, \quad P \left. \frac{\partial R_6(u; \kappa_6)}{\partial u} \right|_{u=0} = h_6(\kappa_6). \quad (4.7)$$

All quantum spectral derivatives are taken at fixed  $\kappa_6$ . The identities in Eq. (4.3) also give

$$R_6(u; \kappa_6)R_6(-u; \kappa_6) = (1 - 4u^2)\mathbf{1}_4. \quad (4.8)$$

This algebraic inversion relation ensures that the inverse required by the finite-lattice time operator exists for  $u \neq \pm \frac{1}{2}$ .

To fix the continuum scaling, consider the lower symbol of a bond Hamiltonian. The coherent states and the smooth unit spin field  $\mathbf{S}(x, t)$  are those of Section 2.2. We write the transverse components as

$$S^\pm := S^1 \pm iS^2, \quad S_n^\pm := S^\pm(x_n, t). \quad (4.9)$$

With the complex extension described in Section 2.2, the unit-spin constraint reads  $S^+S^- + (S^3)^2 = 1$ . Direct evaluation of Eq. (4.1) gives

$$h_{6;n,n+1}^\downarrow = 1 + \mathbf{S}_n^\top \mathbf{S}_{n+1} + \frac{\kappa_6}{2} (S_n^+ S_{n+1}^3 + S_n^3 S_{n+1}^+). \quad (4.10)$$

After subtracting the constant 2, the exchange term starts at order  $\epsilon^2$  by Eq. (2.30), whereas the deformation term tends to  $\kappa_6 S^+ S^3$  for coincident spins. Both terms therefore survive in the same continuum limit if the *Class 6 continuum deformation coupling*  $\alpha_6$  and the continuum spectral parameter  $\lambda \neq 0$  are held fixed under

$$\kappa_6 = \epsilon^2 \alpha_6, \quad u = \frac{\lambda}{\epsilon}, \quad t = \epsilon^2 t_{\text{lat}}, \quad \lambda \neq 0. \quad (4.11)$$

Here  $\epsilon$  is the lattice spacing,  $u$  and  $\lambda$  are the quantum and continuum spectral parameters, and  $t_{\text{lat}}$  and  $t$  are the lattice and continuum times. The deformation scaling agrees with Appendix A.1 of Ref. [7]; the spectral scaling ensures that the normalized spatial Lax operator approaches the identity. If lattice parameters are dimensionless,  $\lambda$  has the dimension of length and  $\alpha_6$  has the dimension of inverse length squared.

In the fundamental representation, define  $L_{6;a,n}(u) := R_{6;a,n}(u; \kappa_6)$  on  $\mathcal{V}_a \otimes \mathcal{H}_n$  and normalize it as

$$\hat{L}_{6;a,n}(\epsilon, \lambda) := \frac{1}{2u} L_{6;a,n}(u). \quad (4.12)$$

Substituting Eq. (4.11) into Eq. (4.5) yields the exact finite expansion

$$\hat{L}_{6;a,n} = \mathbf{1}_{\mathcal{V}_a \otimes \mathcal{H}_n} + \epsilon L_{6;a,n}^{(1)} + \epsilon^2 L_{6;a,n}^{(2)} + \epsilon^3 L_{6;a,n}^{(3)}. \quad (4.13)$$

The superscript labels the power of  $\epsilon$ , and each coefficient is the copy on  $\mathcal{V}_a \otimes \mathcal{H}_n$  of the two-site operator

$$L_6^{(1)} = \frac{P}{2\lambda} + \alpha_6 \lambda K_6 + \alpha_6^2 \lambda^3 K_6^2, \quad (4.14)$$

$$L_6^{(2)} = \frac{\alpha_6}{2} K_6 + \alpha_6^2 \lambda^2 K_6^2, \quad L_6^{(3)} = \frac{\alpha_6^2 \lambda}{4} K_6^2. \quad (4.15)$$

In particular, the operator square of the first coefficient satisfies

$$(L_6^{(1)})^2 = \frac{\mathbf{1}_4}{4\lambda^2} + 2L_6^{(2)}. \quad (4.16)$$

The second coefficient is therefore already present in products of the leading spatial coefficients. It will enter the coherent-state product correction in the time Lax equation.

For the lower-symbol evaluation at  $x = x_n$ , the permutation gives

$$P_{a,n}^\downarrow = \frac{1}{2} \begin{pmatrix} 1 + S^3 & S^- \\ S^+ & 1 - S^3 \end{pmatrix}, \quad (4.17)$$

while the deformation operators give

$$K_{6;a,n}^\downarrow = \begin{pmatrix} S^+/2 & S^3 \\ 0 & -S^+/2 \end{pmatrix}, \quad (K_{6;a,n}^2)^\downarrow = \begin{pmatrix} 0 & -S^+ \\ 0 & 0 \end{pmatrix}. \quad (4.18)$$

The second matrix is obtained by squaring the quantum operator before taking its lower symbol. Using these matrices in Eq. (4.14), the continuum spatial Lax matrix is

$$\begin{aligned} U_6(x, t; \lambda) &:= (L_{6;a,n}^{(1)})^\downarrow|_{x_n=x} \\ &= \frac{1}{4\lambda} \begin{pmatrix} 1 + S^3 + 2\alpha_6\lambda^2 S^+ & S^- + 4\alpha_6\lambda^2 S^3 - 4\alpha_6^2\lambda^4 S^+ \\ S^+ & 1 - S^3 - 2\alpha_6\lambda^2 S^+ \end{pmatrix}. \end{aligned} \quad (4.19)$$

It obeys

$$\widehat{L}_{6;a,n}^\downarrow = \mathbf{1}_{\mathcal{V}_a} + \epsilon U_6(x_n, t; \lambda) + \mathcal{O}(\epsilon^2), \quad \text{tr } U_6 = \frac{1}{2\lambda}. \quad (4.20)$$

We retain the complete  $\mathfrak{gl}(2)$  matrix in the finite-lattice limit and denote its traceless part by

$$U_{6,0} := U_6 - \frac{1}{4\lambda} \mathbf{1}_2. \quad (4.21)$$

At  $\alpha_6 = 0$ , Eq. (4.19) reduces to  $(\mathbf{1}_2 + S^i \sigma^i)/(4\lambda)$ , the spatial Lax matrix of the isotropic Landau–Lifshitz model in the normalization used here. For the deformed model, Eqs. (4.13)–(4.16) provide the operator coefficients needed to construct the time component.

## 4.2 Time Lax matrix from the finite-lattice construction

We apply the finite-lattice construction of Eq. (2.19) to the Class 6 operators of Section 4.1. As in the Class 5 calculation, we remove a field-independent scalar term by choosing

$$\widehat{A}_{6;a,n} := A_{6;a,n} + \frac{\mathbf{i}}{u} \mathbf{1}, \quad (4.22)$$

where the identity acts on the auxiliary space and the physical sites  $n-1, n$ . The normalized spatial operator then gives

$$\widehat{A}_{6;a,n} = -\mathbf{i} \widehat{L}_{6;a,n}^{-1} \left( [h_{6;n-1,n}, \widehat{L}_{6;a,n}] + \partial_u \widehat{L}_{6;a,n} \right). \quad (4.23)$$

The derivative is taken at fixed lattice coupling  $\kappa_6$ . With the scaling in Eq. (4.11), the exact expansion (4.13) therefore implies

$$\partial_u \widehat{L}_{6;a,n} = \epsilon^2 \partial_\lambda L_{6;a,n}^{(1)} + \epsilon^3 \partial_\lambda L_{6;a,n}^{(2)} + \epsilon^4 \partial_\lambda L_{6;a,n}^{(3)}. \quad (4.24)$$

Here and below,  $\partial_\lambda$  holds  $\alpha_6$  fixed, and also holds the spin fixed when acting on a classical matrix.

The bond Hamiltonian is

$$h_{6;n-1,n} = h_{n-1,n}^{(0)} + \epsilon^2 \alpha_6 K_{6;n-1,n}, \quad h^{(0)} := 2P. \quad (4.25)$$

Expanding Eq. (4.23) gives

$$\widehat{A}_{6;a,n} = \epsilon A_{6;a,n}^{(1)} + \epsilon^2 A_{6;a,n}^{(2)} + \mathcal{O}(\epsilon^3), \quad (4.26)$$

where

$$A_{6;a,n}^{(1)} = -\mathbf{i} [h_{n-1,n}^{(0)}, L_{6;a,n}^{(1)}], \quad (4.27)$$

$$\begin{aligned} A_{6;a,n}^{(2)} &= \mathbf{i} L_{6;a,n}^{(1)} [h_{n-1,n}^{(0)}, L_{6;a,n}^{(1)}] \\ &\quad - \mathbf{i} [h_{n-1,n}^{(0)}, L_{6;a,n}^{(2)}] - \mathbf{i} \partial_\lambda L_{6;a,n}^{(1)}. \end{aligned} \quad (4.28)$$

The deformation of the bond Hamiltonian first contributes explicitly at order  $\epsilon^3$ , since its commutator with the leading identity in  $\hat{L}_6$  vanishes. The dependence on  $\alpha_6$  in Eqs. (4.27) and (4.28) enters through  $L_6^{(1)}$  and  $L_6^{(2)}$ .

Taking the lower symbol on the two physical sites gives

$$(A_{6;a,n}^{(1)})^\downarrow = -2(\mathbf{S}_{n-1} \times \mathbf{S}_n)_i \frac{\partial U_6}{\partial S^i}. \quad (4.29)$$

The Cartesian derivatives in this formula act on the explicit matrix (4.19), with  $S^\pm = S^1 \pm iS^2$ , before imposing the unit-spin constraint. Since  $U_6$  is affine in the spin components, these derivatives are independent of the spin. At  $x = x_n$ ,

$$\mathbf{S}_{n-1} \times \mathbf{S}_n = \epsilon \mathbf{S} \times \partial_x \mathbf{S} + \mathcal{O}(\epsilon^2). \quad (4.30)$$

The second time coefficient is needed only at coincident spin arguments. Using  $S^+ S^- + (S^3)^2 = 1$  and the traceless spatial matrix  $U_{6,0}$  of Eq. (4.21), we obtain

$$(A_{6;a,n}^{(2)})^\downarrow \Big|_{\mathbf{S}_{n-1}=\mathbf{S}_n=\mathbf{S}} = -2i\partial_\lambda U_{6,0} + i\alpha_6 S^+ S^3 \mathbf{1}_{\mathcal{V}_a}. \quad (4.31)$$

Consequently,

$$\hat{A}_{6;a,n}^\downarrow = \epsilon^2 V_6^\downarrow(x_n, t; \lambda) + \mathcal{O}(\epsilon^3), \quad (4.32)$$

with

$$V_6^\downarrow = -2(\mathbf{S} \times \partial_x \mathbf{S})_i \frac{\partial U_6}{\partial S^i} - 2i\partial_\lambda U_{6,0} + i\alpha_6 S^+ S^3 \mathbf{1}_{\mathcal{V}_a}. \quad (4.33)$$

The products in the discrete zero-curvature equation require a separate calculation. Let  $\hat{\mathcal{D}}_{6;a,n}^{(\pm)}$  denote Eqs. (2.45) and (2.46) evaluated for Class 6. For the first product, the operator expansions give

$$\begin{aligned} \hat{\mathcal{D}}_{6;a,n}^{(+)} &= \epsilon^2 \left[ (A_{6;a,n+1}^{(1)} L_{6;a,n}^{(1)})^\downarrow - (A_{6;a,n+1}^{(1)})^\downarrow (L_{6;a,n}^{(1)})^\downarrow \right] \\ &\quad + \epsilon^3 \left[ (A_{6;a,n+1}^{(2)} L_{6;a,n}^{(1)} + A_{6;a,n+1}^{(1)} L_{6;a,n}^{(2)})^\downarrow \right. \\ &\quad \left. - (A_{6;a,n+1}^{(2)})^\downarrow (L_{6;a,n}^{(1)})^\downarrow - (A_{6;a,n+1}^{(1)})^\downarrow (L_{6;a,n}^{(2)})^\downarrow \right] + \mathcal{O}(\epsilon^4). \end{aligned} \quad (4.34)$$

The expression for  $\hat{\mathcal{D}}_{6;a,n}^{(-)}$  has the spatial factor on the left and the time factor at  $n$ , supported on  $n-1, n$ . Terms involving the third time coefficient cancel within each difference, because they multiply the leading identity in  $\hat{L}_6$ . The coefficient  $L_6^{(3)}$  first enters these differences at order  $\epsilon^4$ . The continuum source therefore requires the products displayed in Eq. (4.34), including those involving  $L_6^{(2)}$ .

At coincident spins, the order- $\epsilon^2$  terms cancel between  $\hat{\mathcal{D}}_{6;a,n}^{(+)}$  and  $\hat{\mathcal{D}}_{6;a,n}^{(-)}$ . To express the surviving contribution, define  $\Xi_6$  as the difference between the lower symbol of the square of the first spatial coefficient and the square of its lower symbol:

$$\begin{aligned} \Xi_6 &:= \left[ (L_6^{(1)})^2 \right]^\downarrow - U_6^2 \\ &= \frac{1}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a} + 2(L_6^{(2)})^\downarrow - U_6^2. \end{aligned} \quad (4.35)$$

The second equality follows from Eq. (4.16). In particular, Eq. (4.18) gives

$$(L_6^{(2)})^\downarrow = \begin{pmatrix} \alpha_6 S^+ / 4 & \alpha_6 S^3 / 2 - \alpha_6^2 \lambda^2 S^+ \\ 0 & -\alpha_6 S^+ / 4 \end{pmatrix}. \quad (4.36)$$

The first bracket in Eq. (4.34) is  $\mathcal{D}_{6,2}^{(+)}$ , and the second is  $\mathcal{D}_{6,3}^{(+)}$ , with the two spin arguments understood as in Eq. (2.48). The corresponding left-ordered products give  $\mathcal{D}_{6,2}^{(-)}$  and  $\mathcal{D}_{6,3}^{(-)}$ . Applying Eq. (2.34) at the shared physical site yields

$$\mathcal{D}_{6,2}^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_{6,2}^{(-)}(\mathbf{S}, \mathbf{S}) = 2i\Xi_6, \quad (4.37)$$

$$\partial_x S^i \left( \frac{\partial \mathcal{D}_{6,2}^{(+)}}{\partial S_R^i} + \frac{\partial \mathcal{D}_{6,2}^{(-)}}{\partial S_L^i} \right) \Big|_{\mathbf{S}_L = \mathbf{S}_R = \mathbf{S}} = 2i\partial_x \Xi_6, \quad (4.38)$$

$$\mathcal{D}_{6,3}^{(+)}(\mathbf{S}, \mathbf{S}) - \mathcal{D}_{6,3}^{(-)}(\mathbf{S}, \mathbf{S}) = 2i[\Xi_6, U_6]. \quad (4.39)$$

The first identity cancels the order- $\epsilon^2$  difference. The remaining identities display separately the two contributions in Eq. (2.50): the spatial variation supplies the derivative, and the cubic products at coincident spins supply the commutator. Consequently,

$$\mathcal{C}_6 := \lim_{\epsilon \rightarrow 0} \epsilon^{-3} \left( \widehat{\mathcal{D}}_{6;a,n}^{(+)} - \widehat{\mathcal{D}}_{6;a,n}^{(-)} \right) = 2i(\partial_x \Xi_6 + [\Xi_6, U_6]). \quad (4.40)$$

This calculation uses the unit-spin constraint and its spatial derivative and holds for arbitrary time dependence of the spin.

The local time correction

$$\begin{aligned} \Delta V_6 &:= 2i\Xi_6 - \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a} \\ &= 4i(L_6^{(2)})^\downarrow - 2iU_6^2 + \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a} \end{aligned} \quad (4.41)$$

then satisfies

$$\partial_x \Delta V_6 + [\Delta V_6, U_6] = \mathcal{C}_6. \quad (4.42)$$

The field-independent scalar in Eq. (4.41) fixes a representative with traceless final time matrix. To simplify it, the explicit spatial matrix gives

$$\begin{aligned} \det U_6 &= -\frac{\alpha_6}{2} S^+ S^3, \\ \partial_\lambda U_{6,0} &= 4(L_6^{(2)})^\downarrow - \frac{1}{\lambda} U_{6,0}. \end{aligned} \quad (4.43)$$

The determinant identity uses the unit-spin constraint. The Cayley–Hamilton identity, together with  $\text{tr } U_6 = 1/(2\lambda)$ , now reduces Eq. (4.41) to

$$\Delta V_6 = i\partial_\lambda U_{6,0} - i\alpha_6 S^+ S^3 \mathbf{1}_{\mathcal{V}_a}. \quad (4.44)$$

Adding this correction to Eq. (4.33) yields

$$V_6 := V_6^\downarrow + \Delta V_6 = -2(\mathbf{S} \times \partial_x \mathbf{S})^i \frac{\partial U_6}{\partial S^i} - i\partial_\lambda U_{6,0}, \quad \text{tr } V_6 = 0. \quad (4.45)$$

We write  $S_x^a := \partial_x S^a$  for  $a \in \{+, -, 3\}$ . The matrix elements are

$$\begin{aligned} (V_6)_{11} &= \frac{i}{4\lambda^2} \left[ S^3 - 2\alpha_6 \lambda^2 S^+ + \lambda(S^- S_x^+ - S^+ S_x^-) \right. \\ &\quad \left. + 4\alpha_6 \lambda^3 (S^+ S_x^3 - S^3 S_x^+) \right], \end{aligned} \quad (4.46)$$

$$\begin{aligned} (V_6)_{12} &= \frac{i}{4\lambda^2} \left[ S^- - 4\alpha_6 \lambda^2 S^3 + 12\alpha_6^2 \lambda^4 S^+ \right. \\ &\quad \left. + 2\lambda(S^3 S_x^- - S^- S_x^3) + 4\alpha_6 \lambda^3 (S^- S_x^+ - S^+ S_x^-) \right. \\ &\quad \left. + 8\alpha_6^2 \lambda^5 (S^3 S_x^+ - S^+ S_x^3) \right], \end{aligned} \quad (4.47)$$

$$(V_6)_{21} = \frac{i}{4\lambda^2} \left[ S^+ + 2\lambda(S^+ S_x^3 - S^3 S_x^+) \right], \quad (4.48)$$

$$(V_6)_{22} = -(V_6)_{11}. \quad (4.49)$$

The role of the second spatial coefficient is also visible directly in the source. Expanding Eq. (4.40) gives

$$\mathcal{C}_6 = -2i\partial_x(U_6^2) + 4i\partial_x(L_6^{(2)})^\downarrow + 4i[(L_6^{(2)})^\downarrow, U_6]. \quad (4.50)$$

At a fixed time, consider the spatially uniform spin configuration  $\mathbf{S} = (0, 0, 1)^\top$ . Then

$$U_6 = \begin{pmatrix} 1/(2\lambda) & \alpha_6\lambda \\ 0 & 0 \end{pmatrix}, \quad \mathcal{C}_6 = \begin{pmatrix} 0 & -i\alpha_6/\lambda \\ 0 & 0 \end{pmatrix}. \quad (4.51)$$

The source is nonzero for  $\alpha_6 \neq 0$ . Omitting the term involving  $L_6^{(2)}$  from Eq. (4.41) would leave a polynomial in  $U_6$ , whose spatial derivative and commutator with  $U_6$  both vanish in this example. The total-derivative correction sufficient for Class 5 therefore fails for Class 6.

In Section 4.3 we derive the Hamiltonian equations independently and establish their off-shell equivalence to the zero-curvature equation for  $(U_6, V_6)$ .

### 4.3 Zero-curvature equation and the Landau–Lifshitz equations of motion

We derive the continuum Hamiltonian flow from the Class 6 bond Hamiltonian and compare it with the corrected Lax pair of Section 4.2. The Hamiltonian also fixes the continuum limit of the Heisenberg derivative on the left-hand side of the lattice zero-curvature equation.

With  $\mathbf{S}_n(t) = \mathbf{S}(x_n, t)$  and  $\kappa_6 = \epsilon^2\alpha_6$ , expanding the bond lower symbol in Eq. (4.10) gives

$$h_{6;n,n+1}^\downarrow = 2 + \epsilon^2\mathcal{H}_6(x_n, t) + \mathcal{O}(\epsilon^3), \quad (4.52)$$

where the continuum Hamiltonian density at  $(x, t)$  is

$$\mathcal{H}_6(x, t) = -\frac{1}{2}\partial_x\mathbf{S}^\top\partial_x\mathbf{S} + \alpha_6 S^+ S^3. \quad (4.53)$$

Here we used the spatial derivatives of the unit-spin constraint in Eq. (2.30). The anisotropy can be written in terms of the constant symmetric matrix

$$N_6 := \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & -i \\ -1 & -i & 0 \end{pmatrix}, \quad N_6^\top = N_6, \quad N_6^3 = 0. \quad (4.54)$$

Since  $\mathbf{S}^\top N_6 \mathbf{S} = -2S^+ S^3$ , the continuum Hamiltonian at fixed circumference  $\ell = N\epsilon$  is

$$\begin{aligned} H_6^{\text{cl}} &:= \lim_{\epsilon \rightarrow 0} \frac{(H_6^{\text{lat}})^\downarrow - 2N}{\epsilon} \\ &= \int_0^\ell dx \left[ -\frac{1}{2}\partial_x\mathbf{S}^\top\partial_x\mathbf{S} - \frac{\alpha_6}{2}\mathbf{S}^\top N_6 \mathbf{S} \right]. \end{aligned} \quad (4.55)$$

We use the equal-time spin Poisson bracket with the same normalization as in Eq. (3.50),

$$\{S^i(x), S^j(y)\}_6 = 2\varepsilon_{ijk}S^k(x)\delta(x-y), \quad \varepsilon_{123} = 1, \quad (4.56)$$

where  $\varepsilon_{ijk}$  is completely antisymmetric and  $\delta(x-y)$  is the Dirac delta function. The common time argument is suppressed in Eq. (4.56). We take the functional derivative with respect to the Cartesian spin components before imposing  $\mathbf{S}^\top\mathbf{S} = 1$ . Integration by parts with periodic variations then gives  $\delta H_6^{\text{cl}}/\delta\mathbf{S} = \partial_x^2\mathbf{S} - \alpha_6 N_6 \mathbf{S}$ . The Hamiltonian equation  $\partial_t S^i = \{S^i, H_6^{\text{cl}}\}_6$  therefore reads

$$\partial_t \mathbf{S} = -2\mathbf{S} \times \left( \partial_x^2 \mathbf{S} - \alpha_6 N_6 \mathbf{S} \right). \quad (4.57)$$

The cross product is bilinear, as in Section 3.3, and the right-hand side preserves  $\mathbf{S}^\top \mathbf{S} = 1$ . In terms of  $S^\pm$  and  $S^3$ , Eq. (4.57) becomes

$$\partial_t S^+ = 2i \left( S^+ \partial_x^2 S^3 - S^3 \partial_x^2 S^+ + \alpha_6 (S^+)^2 \right), \quad (4.58)$$

$$\partial_t S^- = 2i \left( S^3 \partial_x^2 S^- - S^- \partial_x^2 S^3 + 2\alpha_6 (S^3)^2 - \alpha_6 S^+ S^- \right), \quad (4.59)$$

$$\partial_t S^3 = i \left( S^- \partial_x^2 S^+ - S^+ \partial_x^2 S^- - 2\alpha_6 S^+ S^3 \right). \quad (4.60)$$

The relation to the quantum time evolution can also be checked at finite lattice spacing. Let  $\sigma_n^i$  denote the Pauli matrix  $\sigma^i$  acting at site  $n$ , with identity operators on the other sites, and write  $\boldsymbol{\sigma}_n = (\sigma_n^1, \sigma_n^2, \sigma_n^3)^\top$ . Taking the lower symbol of the Heisenberg derivative componentwise on a product of coherent states at a fixed lattice time gives

$$\begin{aligned} \frac{1}{\epsilon^2} (\partial_{t_{\text{lat}}} \boldsymbol{\sigma}_n)^\downarrow = & -2\mathbf{S}_n \times \left[ \frac{\mathbf{S}_{n-1} + \mathbf{S}_{n+1} - 2\mathbf{S}_n}{\epsilon^2} \right. \\ & \left. - \frac{\alpha_6}{2} N_6 (\mathbf{S}_{n-1} + \mathbf{S}_{n+1}) \right]. \end{aligned} \quad (4.61)$$

This identity follows from the two bonds adjacent to site  $n$  and the Pauli commutation relations. Its smooth-field limit is Eq. (4.57), with the time convention  $t = \epsilon^2 t_{\text{lat}}$ . Applying the same commutator to the normalized spatial operator gives

$$(\partial_{t_{\text{lat}}} \widehat{L}_{6;a,n})^\downarrow = \epsilon^3 \partial_t U_6(x_n, t; \lambda) + \mathcal{O}(\epsilon^4), \quad (4.62)$$

where  $\partial_t \mathbf{S}$  is given by Eq. (4.57). Together with the correction identity (4.42), this supplies both sides of the continuum zero-curvature equation.

To establish equivalence with the equations of motion, we now treat the time derivatives as independent variables. Define the equation-of-motion residual vector

$$\mathbf{E}^{(6)} := \partial_t \mathbf{S} + 2\mathbf{S} \times \left( \partial_x^2 \mathbf{S} - \alpha_6 N_6 \mathbf{S} \right). \quad (4.63)$$

Write  $E_i^{(6)}$  for its Cartesian components and  $E_\pm^{(6)} := E_1^{(6)} \pm iE_2^{(6)}$  for its transverse components. The curvature of the matrices in Eqs. (4.19) and (4.45) is

$$F_{tx}^{(6)} := \partial_t U_6 - \partial_x V_6 + [U_6, V_6]. \quad (4.64)$$

Using only  $\mathbf{S}^\top \mathbf{S} = 1$ ,  $\mathbf{S}^\top \partial_x \mathbf{S} = 0$ , and  $\mathbf{S}^\top \partial_x^2 \mathbf{S} = -\partial_x \mathbf{S}^\top \partial_x \mathbf{S}$ , we obtain the off-shell identity

$$F_{tx}^{(6)} = \frac{1}{4\lambda} \begin{pmatrix} E_3^{(6)} + 2\alpha_6 \lambda^2 E_+^{(6)} & E_-^{(6)} + 4\alpha_6 \lambda^2 E_3^{(6)} - 4\alpha_6^2 \lambda^4 E_+^{(6)} \\ E_+^{(6)} & -E_3^{(6)} - 2\alpha_6 \lambda^2 E_+^{(6)} \end{pmatrix}. \quad (4.65)$$

For  $\lambda \neq 0$ , the inverse relations are

$$E_+^{(6)} = 4\lambda (F_{tx}^{(6)})_{21}, \quad (4.66)$$

$$E_3^{(6)} = 4\lambda \left[ (F_{tx}^{(6)})_{11} - 2\alpha_6 \lambda^2 (F_{tx}^{(6)})_{21} \right], \quad (4.67)$$

$$\begin{aligned} E_-^{(6)} = 4\lambda \left[ (F_{tx}^{(6)})_{12} - 4\alpha_6 \lambda^2 (F_{tx}^{(6)})_{11} \right. \\ \left. + 12\alpha_6^2 \lambda^4 (F_{tx}^{(6)})_{21} \right]. \end{aligned} \quad (4.68)$$

Thus  $F_{tx}^{(6)} = 0$  is equivalent to all three component equations (4.58)–(4.60) for every  $\lambda \neq 0$  and every  $\alpha_6$ . This equivalence is a local identity on the unit-spin constraint surface.

#### 4.4 Comparison with the known eleven-vertex Landau–Lifshitz pair

Class 6 is the eleven-vertex model at the quantum level [8]. The corresponding rational Landau–Lifshitz hierarchy is known [9]; we use the explicit rational Lax pair in Ref. [10], Eqs. (4.3)–(4.5), for comparison. Introduce the traceless spin matrix

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & -\Sigma_{11} \end{pmatrix} \quad (4.69)$$

and spectral parameter  $z$ . The spatial matrix is

$$U_{11v}(z, \Sigma) = \begin{pmatrix} \Sigma_{11}/z - z\Sigma_{12} & \Sigma_{12}/z \\ \Sigma_{21}/z - 2z\Sigma_{11} - z^3\Sigma_{12} & -\Sigma_{11}/z + z\Sigma_{12} \end{pmatrix}. \quad (4.70)$$

Let  $\Lambda_\Sigma \neq 0$  fix the two eigenvalues of the spin matrix through  $\Sigma^2 = \Lambda_\Sigma^2 \mathbf{1}_2$ . Let  $k$  denote the coefficient of the spatial derivative in the zero-curvature equation and let  $\tau$  denote the time variable of the known model. Define

$$M_{\text{top}}(z, \Sigma) = \begin{pmatrix} \Sigma_{12} & 0 \\ 2\Sigma_{11} + 2z^2\Sigma_{12} & -\Sigma_{12} \end{pmatrix}, \quad v = -\frac{k}{4\Lambda_\Sigma^2}[\Sigma, \partial_x \Sigma]. \quad (4.71)$$

The time matrix is

$$V_{11v}(z, \Sigma) = \frac{1}{2} \left[ \frac{1}{z} U_{11v}(z, \Sigma) + 2M_{\text{top}}(z, \Sigma) - U_{11v}(z, v) \right], \quad (4.72)$$

with  $\partial_\tau U_{11v} + k\partial_x V_{11v} - [U_{11v}, V_{11v}] = 0$ .

For the Class 6 spin field, set

$$\Sigma := \begin{pmatrix} S^3 & S^+ \\ S^- & -S^3 \end{pmatrix}. \quad (4.73)$$

The unit-spin constraint gives  $\Sigma^2 = \mathbf{1}_2$ . Let  $\beta \neq 0$  satisfy

$$\beta^2 = -2\alpha_6, \quad z = \beta\lambda, \quad k = -\frac{4}{\beta}, \quad \tau = \frac{i\beta^2}{2}t = -i\alpha_6 t. \quad (4.74)$$

This dictionary applies for  $\alpha_6 \neq 0$ . The Class 6 matrices in Sections 4.1 and 4.2 are regular at  $\alpha_6 = 0$ , and the undeformed limit is taken directly in those expressions. Then the traceless spatial matrix  $U_{6,0}$  defined in Eq. (4.21) and the time matrix Eq. (4.45) satisfy

$$U_{6,0}(\lambda) = \frac{\beta}{4} U_{11v}(\beta\lambda, \Sigma)^\top, \quad (4.75)$$

$$V_6(\lambda) = \frac{i\beta^2}{2} V_{11v}(\beta\lambda, \Sigma)^\top. \quad (4.76)$$

The scalar difference  $U_6 - U_{6,0} = \mathbf{1}_2/(4\lambda)$  does not contribute to the curvature. The transpose reverses the commutator order, while  $(\beta/4)k = -1$  and Eq. (4.74) give the zero-curvature convention used here. The same dictionary maps the eleven-vertex spin equation to Eq. (4.57).

## 5 Comparison of Class 5 and Class 6

### 5.1 Operator-product corrections in the coherent-state limit

Let  $m \in \{5, 6\}$  label the model and define

$$\Xi_m := \left[ (L_m^{(1)})^2 \right]^\downarrow - U_m^2, \quad U_m = (L_m^{(1)})^\downarrow. \quad (5.1)$$

All lower symbols are evaluated at the same spin  $\mathbf{S}(x, t)$ . Sections 3.2 and 4.2 give

$$\mathcal{C}_m = 2i(\partial_x \Xi_m + [\Xi_m, U_m]), \quad (5.2)$$

with the representatives

$$\Delta V_m = 2i\Xi_m - \frac{i}{4\lambda^2} \mathbf{1}_{\mathcal{V}_a}, \quad V_m = V_m^\downarrow + \Delta V_m. \quad (5.3)$$

The scalar term does not affect the curvature.

## 5.2 Deformation scalings and second moments

The different deformation scalings follow from the bond lower symbols. For Class 5, the deformation vanishes at coincident spins and its leading term is proportional to  $\kappa_5 \epsilon$ . For Class 6, it contributes already at order  $\kappa_6$ . Matching these terms to the order- $\epsilon^2$  exchange energy gives  $\kappa_5 = \epsilon \alpha_5$  and  $\kappa_6 = \epsilon^2 \alpha_6$ .

Table 1 compares the Hamiltonian densities and spatial-operator coefficients. Here  $L_5^{(2)} := 0$  denotes the absent second-order coefficient in Eq. (3.13), and  $M_5$  and  $N_6$  are defined in Eqs. (3.46) and (4.54).

|                                             | Class 5                                                        | Class 6                                               |
|---------------------------------------------|----------------------------------------------------------------|-------------------------------------------------------|
| Lattice coupling                            | $\epsilon \alpha_5$                                            | $\epsilon^2 \alpha_6$                                 |
| Second spatial coefficient                  | 0                                                              | $\frac{\alpha_6}{2} K_6 + \alpha_6^2 \lambda^2 K_6^2$ |
| Operator square $(L_m^{(1)})^2$             | $\frac{\mathbf{1}_4}{4\lambda^2}$                              | $\frac{\mathbf{1}_4}{4\lambda^2} + 2L_6^{(2)}$        |
| Deformation term in the Hamiltonian density | $\frac{\alpha_5}{2} \mathbf{S}^\top M_5 \partial_x \mathbf{S}$ | $-\frac{\alpha_6}{2} \mathbf{S}^\top N_6 \mathbf{S}$  |

Table 1: Comparison in the fixed exchange and time normalizations of Sections 3.1 and 4.1. The Hamiltonian densities share the exchange term  $-\frac{1}{2} \partial_x \mathbf{S}^\top \partial_x \mathbf{S}$ . The Class 5 total derivative has been removed using periodicity, as in Eq. (3.49).

The antisymmetric matrix  $M_5$  couples the spin to its spatial derivative in Class 5, whereas the symmetric matrix  $N_6$  gives a local anisotropy term in Class 6. The corresponding second moments are

$$\Xi_5 = \frac{\mathbf{1}_{\mathcal{V}_a}}{4\lambda^2} - U_5^2, \quad \Xi_6 = \frac{\mathbf{1}_{\mathcal{V}_a}}{4\lambda^2} + 2(L_6^{(2)})^\downarrow - U_6^2. \quad (5.4)$$

Thus  $[\Xi_5, U_5] = 0$ , and the Class 5 source is a total spatial derivative. For Class 6, Eq. (4.16) expresses  $\Xi_6$  through  $L_6^{(2)}$ , and Eq. (4.50) contains a nonzero commutator. The uniform configuration Eq. (4.51) isolates this term when all spatial derivatives vanish. In the general formula of Section 2.4, the explicit second-order coefficient cancels after the intermediate products are combined, and Eq. (2.69) reconstructs  $\Xi$  from the leading spatial matrix.

In the undeformed limit, write  $\sigma_a^i$  for the Pauli matrices on the auxiliary space  $\mathcal{V}_a$ . Setting  $\alpha_m = 0$  gives

$$U_m|_{\alpha_m=0} = \frac{\mathbf{1}_{\mathcal{V}_a} + S^i \sigma_a^i}{4\lambda}, \quad (5.5)$$

$$\Delta V_m|_{\alpha_m=0} = -\frac{iS^i \sigma_a^i}{4\lambda^2}, \quad \mathcal{C}_m|_{\alpha_m=0} = -\frac{i(\partial_x S^i) \sigma_a^i}{4\lambda^2}. \quad (5.6)$$

Using  $(S^i \sigma_a^i)^2 = \mathbf{1}_{\mathcal{V}_a}$  on the unit-spin constraint, the corrected time matrix is

$$V_m|_{\alpha_m=0} = -\frac{(\mathbf{S} \times \partial_x \mathbf{S})^i \sigma_a^i}{2\lambda} + \frac{iS^i \sigma_a^i}{4\lambda^2}. \quad (5.7)$$

The direct symbol has twice the coefficient of the second term in Eq. (5.7), so the shared-site correction is also required in the spin- $\frac{1}{2}$  XXX limit.

## 6 Summary and discussion

We have derived the continuum time Lax matrices of the Class 5 and Class 6 Landau–Lifshitz models from their quantum lattice time operators at fixed spin  $\frac{1}{2}$ . In the local lattice zero-curvature equation, the two operator products share a physical site. Their coincident order- $\epsilon^2$  contributions cancel, while the first spatial Taylor terms and the coincident order- $\epsilon^3$  terms remain at the continuum time-evolution order.

For the operator expansion of Section 2.4, the surviving contribution is expressed through  $\Xi = (X^2)^\downarrow - (X^\downarrow)^2$ . With the corresponding Sutherland relation,

$$\mathcal{C} = 2i(\partial_x \Xi + [\Xi, U]), \quad (6.1)$$

and the source is absorbed into a local correction of the time matrix. The Class 5 and Class 6 lattice calculations reproduce this expression and give zero-curvature equations that are off-shell equivalent to the independently derived Hamiltonian equations.

In Class 5,  $[\Xi_5, U_5] = 0$  and the source is a total spatial derivative. In Class 6, the representation of  $\Xi_6$  through  $L_6^{(2)}$  gives a nonzero commutator, which remains on a spatially uniform spin configuration. The explicit second-order spatial coefficient is needed in the intermediate operator products but cancels from the general expression for  $\mathcal{C}$ .

The resulting classical dynamics admit independent comparisons. The Class 5 time matrix matches the classical  $r$ -matrix/monodromy result up to a field-independent scalar matrix. Independently, the field transformation of Section 3.5 maps the Class 5 Hamiltonian density and equations of motion locally to those of the null-like warped  $SL(2)$  Landau–Lifshitz model. With the same field, coupling and time dictionary, the traceless Lax pairs are related by a local auxiliary-space gauge transformation. The transformation is not generically periodic, so the periodic monodromies and global boundary sectors are not identified. For Class 6, the spatial matrix, time matrix and spin equation map to the rational eleven-vertex Landau–Lifshitz pair through the dictionary in Section 4.4. In the undeformed XXX limit, the direct time symbol has twice the correct coefficient of one term; the shared-site correction restores the standard isotropic time matrix.

The numerical searches in Appendix B recover coefficients consistent with the analytic Class 5 and Class 6 pairs at the tested parameter values. The final Lax relations are established by the operator calculation and the off-shell curvature identities.

The derivation is local on the complex unit-spin constraint surface and uses the fixed spin- $\frac{1}{2}$  scaling. Periodicity enters the Hamiltonian variation and the Class 5 monodromy comparison. On the quantum lattice, the comparison uses the algebraic commutator evolution defined in Eq. (2.18) and its instantaneous lower symbols on product coherent states. Finite-time control would require estimates of the correlations generated between sites and of the accumulated continuum error.

For higher lattice Hamiltonians, the time operators act on more physical sites and the overlap terms can involve higher local operator moments. Comparing those flows with the boost-generated charges of Ref. [7] and the eleven-vertex hierarchy of Ref. [9] would test the extension of Eq. (2.67). A separate extension to higher local spin would clarify the relation between the fixed-spin long-wavelength limit used here and a large-spin classical limit.

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literature searches, and organization and editing of the manuscript. The analytic and numerical calculations and the references supporting the scientific claims were checked by the author, who takes responsibility for the content of the paper.

## Data availability

No external experimental or observational data were used in this work. Numerical data generated by the coefficient searches and validation calculations are available in the accompanying repository.

## Code availability

Code for the machine-learning coefficient searches and the symbolic and finite-lattice checks, together with numerical outputs, is available in the accompanying repository:

<https://github.com/ryos-physrockme/llsm-class5-class6-lax>.

Source code is released under the MIT License.

## A Derivation of the shared-site identity

This appendix derives Eqs. (2.63)–(2.66). The assumptions and operator expansions are those of Section 2.4. All lower symbols below are evaluated with the same rank-one spin state  $\rho = (\mathbf{1}_2 + S^i \sigma^i)/2$  on the two physical sites, while the auxiliary-space matrix order is kept fixed.

From Eq. (2.62),

$$A^{(1)} = -2i(X_1 - X_2)P. \quad (\text{A.1})$$

At coincident spins,  $P(\rho \otimes \rho) = (\rho \otimes \rho)P = \rho \otimes \rho$ , so  $\langle A^{(1)} \rangle = 0$ . Direct substitution in the two overlap products gives

$$\mathcal{D}_2^{(+)}(\mathbf{S}, \mathbf{S}) = \mathcal{D}_2^{(-)}(\mathbf{S}, \mathbf{S}) = 2i \left[ (X^2)^\dagger - U^2 \right] = 2i\Xi. \quad (\text{A.2})$$

Thus the coincident order- $\epsilon^2$  contribution cancels in  $\hat{\mathcal{D}}^{(+)} - \hat{\mathcal{D}}^{(-)}$ .

To obtain its first spatial variation, choose at one point a fixed physical basis in which

$$\rho = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \partial_x \rho = \begin{pmatrix} 0 & p \\ q & 0 \end{pmatrix}, \quad X = \sum_{r,s=0}^1 X_{rs} \otimes |r\rangle\langle s|. \quad (\text{A.3})$$

The basis is held fixed when differentiating. The projector identity  $\rho \partial_x \rho + (\partial_x \rho) \rho = \partial_x \rho$  removes the diagonal components of  $\partial_x \rho$ . In this basis

$$U = X_{00}, \quad \Xi = X_{01}X_{10}, \quad (\text{A.4})$$

and differentiation with respect to the spin state gives

$$\partial_x \Xi = p(X_{11} - X_{00})X_{10} + qX_{01}(X_{11} - X_{00}). \quad (\text{A.5})$$

Let  $\delta_R$  and  $\delta_L$  denote directional variations with respect to the right and left physical density matrices, respectively. Their action on the quadratic overlap coefficients is

$$\begin{aligned} \delta_R \mathcal{D}_2^{(+)}[\partial_x \rho] &= 2ip(X_{11} - X_{00})X_{10}, \\ \delta_L \mathcal{D}_2^{(-)}[\partial_x \rho] &= 2iqX_{01}(X_{11} - X_{00}). \end{aligned} \quad (\text{A.6})$$

The minus sign in the definition of the overlap source is compensated by the Taylor expansion of the left neighbour,  $\rho(x - \epsilon) = \rho(x) - \epsilon \partial_x \rho + \dots$ . Therefore the two terms in Eq. (A.6) add and reproduce Eq. (2.64).

It remains to evaluate the coincident order- $\epsilon^3$  term. Using  $\langle A^{(1)} \rangle = 0$ , one finds

$$\mathcal{D}_3^{(+)} - \mathcal{D}_3^{(-)} = \left\langle A^{(2)}X_1 - X_2A^{(2)} + A^{(1)}Y_1 - Y_2A^{(1)} \right\rangle - [\langle A^{(2)} \rangle, U]. \quad (\text{A.7})$$

We separate the four terms in  $A^{(2)}$  according to their origin. The exchange part gives

$$(\mathcal{D}_3^{(+)} - \mathcal{D}_3^{(-)})_{\text{ex}} = 2i[\Xi, U] - i\langle X_1[2P, X_2X_1] \rangle. \quad (\text{A.8})$$

The terms containing the explicit second spatial coefficient cancel only after both its occurrence in  $A^{(2)}$  and the products with  $A^{(1)}$  are retained:

$$\begin{aligned} (\mathcal{D}_3^{(+)} - \mathcal{D}_3^{(-)})_Y &= -2i\langle ([Y_1, X_2] + [X_1, Y_2])P \rangle \\ &= -2i\langle [Y^\dagger, U] + [U, Y^\dagger] \rangle = 0. \end{aligned} \quad (\text{A.9})$$

The spectral-derivative contribution is

$$(\mathcal{D}_3^{(+)} - \mathcal{D}_3^{(-)})_Z = -i\langle X_1(-Z_1 + Z_2) \rangle. \quad (\text{A.10})$$

For the Hamiltonian correction define

$$K_h := \langle [h^{(1)}, X_2] \rangle. \quad (\text{A.11})$$

The triplet condition (2.57) implies

$$\langle (X_1 + X_2)[h^{(1)}, X_1 + X_2] \rangle = 0, \quad (\text{A.12})$$

$$\langle [h^{(1)}, X_2X_1] \rangle = [K_h, U]. \quad (\text{A.13})$$

The first identity follows because  $X_1 + X_2$  commutes with the site permutation and therefore preserves the triplet containing the product state. For the second identity, use the basis of Eq. (A.3). If  $a_h = (h^{(1)})_{00,01}$  and  $b_h = (h^{(1)})_{01,00}$ , the triplet condition gives

$$K_h = a_h X_{10} - b_h X_{01}, \quad \langle [h^{(1)}, X_2X_1] \rangle = a_h [X_{10}, X_{00}] - b_h [X_{01}, X_{00}], \quad (\text{A.14})$$

which is Eq. (A.13). The complete Hamiltonian contribution then reduces to

$$(\mathcal{D}_3^{(+)} - \mathcal{D}_3^{(-)})_h = -i\langle X_1[h^{(1)}, X_1 + X_2] \rangle. \quad (\text{A.15})$$

Adding Eqs. (A.8), (A.9), (A.10), and (A.15), and using the definition (2.56), gives

$$\mathcal{D}_3^{(+)}(\mathcal{S}, \mathcal{S}) - \mathcal{D}_3^{(-)}(\mathcal{S}, \mathcal{S}) = 2i[\Xi, U] - i\langle X_1\mathcal{B} \rangle. \quad (\text{A.16})$$

Together with the Taylor term (2.64), this yields

$$\mathcal{C} = 2i(\partial_x \Xi + [\Xi, U]) - i\langle X_1\mathcal{B} \rangle, \quad (\text{A.17})$$

which is Eq. (2.66). No Cayley–Hamilton relation in the auxiliary space has been used, so the argument allows arbitrary finite auxiliary dimension. The two-dimensional physical spin space enters through the rank-one spin- $\frac{1}{2}$  projector and through Eq. (A.14).

## B Machine-learning searches for the Lax pairs

We determine coefficients in finite Lax ansätze by minimizing an off-shell residual, following the optimization approach of Ref. [14]. The input data are local values of the spin and its derivatives,  $(\mathbf{S}, \partial_x \mathbf{S}, \partial_x^2 \mathbf{S}, \partial_t \mathbf{S})$ . These are sampled subject to the unit-spin constraint and its differentiated identities, with the time derivative independent of the equations of motion. The sampling distributions are specified in Section B.3.

For each model  $m \in \{5, 6\}$ , a run holds the deformation parameter  $\alpha_m$  and spectral parameter  $\lambda$  fixed. The equations of motion and the tensor structures allowed in the Lax ansatz are also supplied. The trainable parameters are the coefficients of those structures: thirteen real coefficients for Class 5 and eight for Class 6, each shared by all samples in a run. They are adjusted so that the curvature agrees with the equation-of-motion residual multiplied by the coefficient map determined from the spatial ansatz. This yields a candidate Lax pair at the chosen  $(\alpha_m, \lambda)$ . Training requires no target values of a Lax pair. The numerical runs use real  $\alpha_m$  and  $\lambda \in i\mathbb{R}_{>0}$ , for which the coefficient relations below are real. Their analytic verification applies to complex  $\lambda \neq 0$ .

To write the ansätze, represent traceless matrices by three-component vectors. With the Pauli matrices acting on the two-dimensional auxiliary space, define

$$\iota(\mathbf{v}) := -\frac{i}{2} v^i \sigma^i, \quad [\iota(\mathbf{v}), \iota(\mathbf{w})] = \iota(\mathbf{v} \times \mathbf{w}). \quad (\text{B.1})$$

Write the candidate matrices as  $U_m^{\text{fit}} = \mu_m \mathbf{1}_2 + \iota(\mathbf{u}_m)$  and  $V_m^{\text{fit}} = \nu_m \mathbf{1}_2 + \iota(\mathbf{v}_m)$ . Here  $\mathbf{u}_m, \mathbf{v}_m$  contain the trainable coefficients, while the scalar parts  $\mu_m, \nu_m$  are prescribed functions, specified below. The superscript distinguishes the fitted candidate from the analytic matrices in the main text. The vector part of its curvature is

$$\mathcal{F}^{(m)} := \partial_t \mathbf{u}_m - \partial_x \mathbf{v}_m + \mathbf{u}_m \times \mathbf{v}_m. \quad (\text{B.2})$$

The vectors  $\mathbf{E}^{(5)}$  and  $\mathbf{E}^{(6)}$  denote  $\partial_t \mathbf{S}$  minus the right-hand side of Eqs. (3.52) and (4.57), respectively. They are computed from the input data at each sampled point. All spin contractions and cross products are complex bilinear. In the losses,  $\|\mathbf{v}\|^2 = \sum_i |v^i|^2$  is the positive Euclidean norm squared, and angle brackets denote the arithmetic mean over the training samples.

### B.1 Class 5

The constant matrix  $M_5$  in Eq. (3.46) can be written as  $M_5 \mathbf{v} = \mathbf{m}_5 \times \mathbf{v}$ , where

$$\mathbf{m}_5 = (i, -1, 0)^\top, \quad \mathbf{m}_5^\top \mathbf{m}_5 = 0, \quad M_5^2 = \mathbf{m}_5 \mathbf{m}_5^\top. \quad (\text{B.3})$$

The ansatz uses the spatial differential operator  $\mathcal{D}_x \mathbf{v} := \partial_x \mathbf{v} + (\alpha_5/2) M_5 \mathbf{v}$  and the constant complex orthogonal matrix

$$O_5(\lambda) := \exp(\alpha_5 \lambda M_5) = \mathbf{1}_3 + \alpha_5 \lambda M_5 + \frac{\alpha_5^2 \lambda^2}{2} M_5^2, \quad O_5^\top O_5 = \mathbf{1}_3, \quad \det O_5 = 1. \quad (\text{B.4})$$

Both  $\mathcal{D}_x$  and  $O_5$  are fixed by  $M_5$ ,  $\alpha_5$  and  $\lambda$  throughout a run.

Define the scalar contractions  $\chi := \mathbf{m}_5^\top \mathbf{S}$ ,  $\xi := \mathbf{m}_5^\top \mathcal{D}_x \mathbf{S}$ , and  $\varrho := \mathbf{m}_5^\top (\mathbf{S} \times \mathcal{D}_x \mathbf{S})$ . The twelve vector structures  $\mathbf{W}_j$  used for the time component are

$$\begin{aligned} \mathbf{W}_1 &= \mathbf{S} \times \mathcal{D}_x \mathbf{S}, & \mathbf{W}_2 &= \varrho \mathbf{m}_5, \\ \mathbf{W}_3 &= \mathbf{S}, & \mathbf{W}_4 &= \chi \mathbf{m}_5, \\ \mathbf{W}_5 &= \mathcal{D}_x \mathbf{S}, & \mathbf{W}_6 &= \chi \mathcal{D}_x \mathbf{S}, \\ \mathbf{W}_7 &= \xi \mathbf{S}, & \mathbf{W}_8 &= \mathbf{m}_5 \times \mathcal{D}_x \mathbf{S}, \\ \mathbf{W}_9 &= \chi (\mathbf{m}_5 \times \mathbf{S}), & \mathbf{W}_{10} &= \varrho \mathbf{S}, \\ \mathbf{W}_{11} &= \mathbf{m}_5, & \mathbf{W}_{12} &= \chi^2 \mathbf{m}_5. \end{aligned} \quad (\text{B.5})$$

The trainable coefficients are the spatial coefficient  $\beta_5$  and the twelve temporal coefficients  $\theta_1, \dots, \theta_{12}$ . The candidate vectors are

$$\mathbf{u}_5 = O_5 \left( \frac{i}{2\lambda} \mathbf{S} + \beta_5 \chi \mathbf{m}_5 \right) - \frac{\alpha_5}{2} \mathbf{m}_5, \quad (\text{B.6})$$

$$\mathbf{v}_5 = -2O_5 \sum_{j=1}^{12} \theta_j \mathbf{W}_j. \quad (\text{B.7})$$

The coefficient of  $\partial_t \mathbf{S}$  in  $\partial_t \mathbf{u}_5$  determines the matrix

$$\mathbf{Q}_5 := O_5 \left( \frac{i}{2\lambda} \mathbf{1}_3 + \beta_5 \mathbf{m}_5 \mathbf{m}_5^\top \right). \quad (\text{B.8})$$

Thus  $\mathbf{Q}_5$  depends on the same  $\beta_5$  as the spatial ansatz. Since  $\mathbf{E}^{(5)}$  contains  $\partial_t \mathbf{S}$  with unit coefficient, the time-derivative terms in  $\mathcal{F}^{(5)} - \mathbf{Q}_5 \mathbf{E}^{(5)}$  cancel for every value of the fitted coefficients. The optimization matches the remaining spatial-derivative and algebraic terms to the supplied dynamics. For a prescribed sparsity weight  $\eta > 0$  and smoothing scale  $\delta = 10^{-10}$ , the loss minimized over these thirteen coefficients is

$$\begin{aligned} \mathcal{L}_5 = & \frac{1}{12} \left\langle \left\| O_5^{-1} (\mathcal{F}^{(5)} - \mathbf{Q}_5 \mathbf{E}^{(5)}) \right\|^2 \right\rangle \\ & + \eta \left( \sqrt{\beta_5^2 + \delta^2} + \sum_{j=1}^{12} \sqrt{\theta_j^2 + \delta^2} \right). \end{aligned} \quad (\text{B.9})$$

The first term measures the curvature mismatch with a fixed, invertible weighting  $O_5^{-1}$  of its components. The second term is a smooth  $\ell^1$  penalty applied uniformly to all thirteen coefficients. The weight  $\eta$  is held fixed during each optimization stage, with the values given in Section B.3.

The scalar parts are fixed using the local conservation law for  $S^+$  in Eq. (3.53). Choosing the scalar representative of the main-text Lax pair gives

$$\begin{aligned} \mu_5 &= \frac{1}{4\lambda} + \frac{\alpha_5}{4} S^+, \\ \nu_5 &= \frac{i\alpha_5}{2} (S^+ \partial_x S^3 - S^3 \partial_x S^+) + \frac{i\alpha_5^2}{4} (S^+)^2. \end{aligned} \quad (\text{B.10})$$

They obey  $\partial_t \mu_5 - \partial_x \nu_5 = (\alpha_5/4) E_+^{(5)}$ , where  $E_+^{(5)}$  is defined in Eq. (3.58). This identity holds independently of the fitted coefficients, so the numerical search determines the traceless parts of the pair.

The numerical coefficients approach the relations

$$\begin{aligned} \beta_5 &= -\frac{i\alpha_5^2 \lambda}{4}, & \theta_1 &= \frac{i}{2\lambda}, & \theta_2 &= \beta_5, \\ \theta_3 &= \frac{1}{4\lambda^2}, & \theta_4 &= \frac{\alpha_5^2}{8}, & \theta_5 &= \dots = \theta_{12} = 0. \end{aligned} \quad (\text{B.11})$$

Substitution gives exactly the matrices in Eqs. (3.18) and (3.41). The identity  $\det \mathbf{Q}_5 = (i/(2\lambda))^3$  and the inverse relations (3.63) establish the local equivalence to the equations of motion.

## B.2 Class 6

For Class 6, the known nilpotency  $N_6^3 = 0$  of Eq. (4.54) determines the finite matrix basis  $\{\mathbf{1}_3, N_6, N_6^2\}$ . The eight real coefficients  $a_1, a_2, b_0, b_1, b_2, c_0, c_1, c_2$  are the trainable parameters. They enter three coefficient matrices,

$$\begin{aligned} \mathbf{A}_6 &= \frac{i}{2\lambda} \mathbf{1}_3 + a_1 N_6 + a_2 N_6^2, \\ \mathbf{B}_6 &= b_0 \mathbf{1}_3 + b_1 N_6 + b_2 N_6^2, \\ \mathbf{C}_6 &= c_0 \mathbf{1}_3 + c_1 N_6 + c_2 N_6^2. \end{aligned} \quad (\text{B.12})$$

The coefficient  $i/(2\lambda)$  of  $\mathbf{1}_3$  in  $\mathbf{A}_6$  is fixed throughout a run. The candidate vectors and scalar parts are

$$\begin{aligned} \mathbf{u}_6 &= \mathbf{A}_6 \mathbf{S}, & \mu_6 &= \frac{1}{4\lambda}, \\ \mathbf{v}_6 &= -2[\mathbf{B}_6 \mathbf{S} + \mathbf{C}_6(\mathbf{S} \times \partial_x \mathbf{S})], & \nu_6 &= 0. \end{aligned} \quad (\text{B.13})$$

Since  $\partial_t \mathbf{u}_6 = \mathbf{A}_6 \partial_t \mathbf{S}$ , the spatial ansatz fixes the coefficient map multiplying  $\mathbf{E}^{(6)}$  to be  $\mathbf{A}_6$ . The loss minimized over the eight coefficients is

$$\mathcal{L}_6 = \frac{1}{12} \left\langle \left\| \mathcal{F}^{(6)} - \mathbf{A}_6 \mathbf{E}^{(6)} \right\|^2 \right\rangle, \quad (\text{B.14})$$

evaluated on the independently sampled spin derivatives. The nilpotency of  $N_6$  gives  $\det \mathbf{A}_6 = (i/(2\lambda))^3$  for every choice of the fitted coefficients, so its coefficient map is invertible for  $\lambda \neq 0$ . As in Class 5, the terms containing  $\partial_t \mathbf{S}$  cancel between  $\mathcal{F}^{(6)}$  and  $\mathbf{A}_6 \mathbf{E}^{(6)}$  for arbitrary fitted coefficients. The sampled time derivative therefore supplies no additional fitting constraint. The constant scalar parts in Eq. (B.13) have identically zero scalar curvature.

The recovered coefficient relations are

$$\begin{aligned} a_1 = c_1 &= -i\alpha_6 \lambda, & a_2 = c_2 &= -i\alpha_6^2 \lambda^3, \\ b_0 &= \frac{1}{4\lambda^2}, & b_1 &= \frac{\alpha_6}{2}, \\ b_2 &= \frac{3\alpha_6^2 \lambda^2}{2}, & c_0 &= \frac{i}{2\lambda}. \end{aligned} \quad (\text{B.15})$$

They give  $\mathbf{C}_6 = \mathbf{A}_6$  and the finite matrix-square-root identity

$$\mathbf{A}_6^2 = -\frac{1}{4\lambda^2} \mathbf{1}_3 + \alpha_6 N_6. \quad (\text{B.16})$$

The corresponding  $U_6^{\text{fit}}, V_6^{\text{fit}}$  coincide with Eqs. (4.19) and (4.45). Their subsequent analytic verification gives  $\mathcal{F}^{(6)} = \mathbf{A}_6 \mathbf{E}^{(6)}$  off shell, with  $\det \mathbf{A}_6 = (i/(2\lambda))^3$ . Thus the final equivalence to the equations of motion follows from an invertible coefficient map, as displayed explicitly in Section 4.3.

### B.3 Sampling and numerical checks

The following distributions generate the input data. All dimensionful quantities are expressed in a fixed length unit. In Class 5, draw a real unit vector  $\mathbf{S}$  by normalizing a three-dimensional standard Gaussian vector. Let  $\Pi_{\mathbf{S}} := \mathbf{1}_3 - \mathbf{S}\mathbf{S}^T$  be its tangent projection, and draw independent standard real Gaussian vectors  $\mathbf{g}_1, \mathbf{g}_2, \mathbf{g}_t$ . The local derivatives are

$$\begin{aligned} \partial_x \mathbf{S} &= \Pi_{\mathbf{S}} \mathbf{g}_1 - \frac{\alpha_5}{2} M_5 \mathbf{S}, \\ \partial_x^2 \mathbf{S} &= \Pi_{\mathbf{S}} \mathbf{g}_2 - (\Pi_{\mathbf{S}} \mathbf{g}_1)^T (\Pi_{\mathbf{S}} \mathbf{g}_1) \mathbf{S} \\ &\quad - \alpha_5 M_5 \Pi_{\mathbf{S}} \mathbf{g}_1 + \frac{\alpha_5^2}{4} M_5^2 \mathbf{S}, \\ \partial_t \mathbf{S} &= -2\Pi_{\mathbf{S}} \mathbf{g}_t. \end{aligned} \quad (\text{B.17})$$

These data satisfy the unit-spin constraint and its differentiated identities; the time derivative is sampled independently of the equations of motion. Each run uses 768 training points and 4096 independent validation points. The pairs  $(\alpha_5, \lambda)$  are  $(0.8, 5i/7)$ ,  $(0.8, 5i/12)$ , and  $(1.4, 5i/12)$ , with three initializations for each pair.

For Class 6, draw a complex Gaussian vector  $\mathbf{z}$ , whose real and imaginary components have independent unit variance, and set  $\mathbf{S} = \mathbf{z}/\sqrt{\mathbf{z}^\top \mathbf{z}}$ . Samples with  $|\mathbf{z}^\top \mathbf{z}| \leq 0.25$  are rejected. Using independent complex Gaussian vectors  $\mathbf{g}_1, \mathbf{g}_2$  with the same distribution, set

$$\begin{aligned}\partial_x \mathbf{S} &= 0.8 \Pi \mathbf{S} \mathbf{g}_1, \\ \partial_x^2 \mathbf{S} &= 0.8 \Pi \mathbf{S} \mathbf{g}_2 - (\partial_x \mathbf{S})^\top (\partial_x \mathbf{S}) \mathbf{S}.\end{aligned}\tag{B.18}$$

For the off-shell loss, an independent complex Gaussian vector  $\mathbf{g}_t$  with the same distribution gives  $\partial_t \mathbf{S} = -2 \Pi \mathbf{S} \mathbf{g}_t$ . Each run uses 768 training points and 3072 independent validation points. The pairs  $(\alpha_6, \lambda)$  are  $(0.35, 10i/13)$ ,  $(0.7, i/2)$ , and  $(1.1, 10i/29)$ , with two initializations each.

Class 5 starts with independent zero-mean Gaussian coefficients of standard deviation 0.25. Class 6 starts from the undeformed values  $b_0 = 1/(4\lambda^2)$ ,  $c_0 = i/(2\lambda)$ , and zero for the remaining coefficients, with independent zero-mean real Gaussian perturbations of standard deviation 0.08. Both searches use Adam followed by the limited-memory Broyden–Fletcher–Goldfarb–Shanno method (L-BFGS). For Class 5, Adam has 600 steps and learning rate 0.03; L-BFGS has at most 50 iterations and learning rate 0.8. The sparsity weight in Eq. (B.9) is  $\eta = 10^{-9}$  during Adam and  $5 \times 10^{-11}$  during L-BFGS. For Class 6, Adam has 1300 steps and learning rate 0.02; L-BFGS has at most 120 iterations and learning rate 0.7. At every step, the loss is evaluated on the full, fixed training set, and its gradients update the ansatz coefficients. Both searches use double precision. After optimization, the same residuals are evaluated on independently generated validation samples; the sparsity penalty is excluded from the Class 5 validation residual.

Across the runs specified above, the largest absolute coefficient error was  $3.61 \times 10^{-7}$  in Class 5 and  $1.12 \times 10^{-11}$  in Class 6. The largest absolute coefficient of  $\mathbf{W}_5, \dots, \mathbf{W}_{12}$  was  $8.10 \times 10^{-8}$ . The maximum validation mean-square residual in the first term of Eq. (B.9) was  $4.94 \times 10^{-14}$ . For Class 6, the maximum validation root-mean-square residual,  $\sqrt{\mathcal{L}_6}$ , was  $2.92 \times 10^{-11}$ . The fixed normalizations and the undeformed Class 6 initial point are inputs to the searches. The deformation-dependent coefficient relations are checked after optimization. Each run fits a finite set of coefficients at one chosen  $(\alpha_m, \lambda)$ ; the general spectral dependence is established by exact substitution into the Lax matrices and the off-shell identities in Sections 3.3 and 4.3. The evaluation of overlapping quantum products in the main text supplies the further connection to the finite lattice time operators.

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